

# Linear Algebra Done Right

## Reading Notes

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# 1 Vector Spaces

## 1.1 $\mathbb{R}^n$ and $\mathbb{C}^n$

Before defining vector spaces, Axler establishes the fields we are working over. The symbol  $\mathbb{F}$  will always denote either the real numbers  $\mathbb{R}$  or the complex numbers  $\mathbb{C}$ .

**Definition 1.1** (Fields). *A field is a set containing at least two distinct elements called 0 and 1, along with operations of addition and multiplication satisfying the properties of commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity.*

### 1.1.1 Complex Numbers, $\mathbb{C}$

**Definition 1.2** (Complex Numbers).

- A complex number is an ordered pair  $(a, b)$  where  $a, b \in \mathbb{R}$ , but we will write this as  $a + bi$ .
- The set of all complex numbers is denoted by  $\mathbb{C}$ :

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

- Addition and multiplication on  $\mathbb{C}$  are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i;$$

here  $a, b, c, d \in \mathbb{R}$

#### My thoughts

Just assume  $i^2 = -1$  and everything works the same as in  $\mathbb{R}$ . Commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity are all properties of complex arithmetic.

### 1.1.2 Lists

**Definition 1.3** (List, length).

- Suppose  $n$  is a nonnegative integer. A list of length  $n$  is an ordered collection of  $n$  elements (which might be numbers, other lists, or more abstract objects).
- Two lists are equal if and only if they have the same length and the same elements in the same order.

#### My thoughts

A list can have multiple same entries unlike sets. Fields and Vector Spaces (defined later) are sets, but a list is a tuple.

### 1.1.3 $\mathbb{F}^n$

**Definition 1.4** ( $\mathbb{F}^n$ , coordinate).  $\mathbb{F}^n$  is the set of all lists of length  $n$  of elements of  $\mathbb{F}$ :

$$\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k \in \{1, \dots, n\}\}$$

For  $(x_1, \dots, x_n) \in \mathbb{F}^n$  and  $k \in \{1, \dots, n\}$ , we say that  $x_k$  is the  $k^{\text{th}}$  coordinate of  $(x_1, \dots, x_n)$ .

**Definition 1.5** (addition in  $F^n$ ). *Addition in  $F^n$  is defined by adding corresponding coordinates:*

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

**Definition 1.6** (additive inverse in  $\mathbb{F}^n$ ,  $-x$ ). *For  $x \in \mathbb{F}^n$ , the additive inverse of  $x$ , denoted by  $-x$ , is the vector  $-x \in \mathbb{F}^n$  such that*

$$x + (-x) = 0$$

*Thus if  $x = (x_1, \dots, x_n)$ , then  $-x = (-x_1, \dots, -x_n)$ .*

**Definition 1.7** (scalar multiplication in  $\mathbb{F}^n$ ). *The product of a number  $\lambda$  and a vector in  $\mathbb{F}^n$  is computed by multiplying each coordinate of the vector by  $\lambda$ :*

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n);$$

*here  $\lambda \in \mathbb{F}$  and  $(x_1, \dots, x_n) \in \mathbb{F}^n$*

## 1.2 Definition of a Vector Space

Now we abstract the properties of  $\mathbb{R}^n$  and  $\mathbb{C}^n$  into a general structure.

**Definition 1.8** (Vector Space). *A vector space is a set  $\mathcal{V}$  along with an addition on  $\mathcal{V}$  and a scalar multiplication on  $\mathcal{V}$  such that commutativity, associativity, identities, additive inverse, and distributivity hold.*

### My thoughts

Essentially, the difference between a field and a vector space is the fact that only scalar multiplication is defined for elements of a vector space. This also implies it is not necessary for a vector space to have multiplicative inverses. Also note, a field is a standalone set while a vector space relies on an underlying field to exist at all.

**Definition 1.9** (Real Vector Space, Complex Vector Space).

- *A vector space over  $\mathbb{R}$  is called a real vector space.*
- *A vector space over  $\mathbb{C}$  is called a complex vector space.*

### 1.2.1 Some Property Proofs

**Theorem 1.10** (Unique Additive Identity). *A vector space has a unique additive identity.*

*Proof.* Suppose  $0$  and  $0'$  are both additive identities for  $V$ . Then

$$0' = 0' + 0 = 0 + 0' = 0,$$

where the first equality holds because  $0$  is an additive identity, the second equality comes from commutativity, and the third equality holds because  $0'$  is an additive identity. Thus  $0' = 0$ , proving that  $\mathcal{V}$  has only one additive identity.  $\square$

**Theorem 1.11.** *a number times the vector  $0$  is  $0$*

*Proof.* For  $v \in \mathcal{V}$ , we have  $0v = (0 + 0)v = 0v + 0v$ . Adding the additive inverse of  $0v$  to both sides of the equation above gives  $0 = 0v$ , as desired.  $\square$

**Theorem 1.12.** *the number  $-1$  times a vector gives the additive inverse of the vector*

*Proof.* For  $v \in \mathcal{V}$ , we have  $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$ . This equation says that  $(-1)v$ , when added to  $v$ , gives  $0$ . Thus  $(-1)v$  is the additive inverse of  $v$ , as desired  $\square$

### 1.3 Subspaces

**Definition 1.13** (Subspace). A subset  $U$  of  $\mathcal{V}$  is called a subspace of  $\mathcal{V}$  if  $U$  is also a vector space under the same addition and scalar multiplication.

#### 1.3.1 Sums of Subspaces

**Definition 1.14** (Sum of Subspaces). Suppose  $\mathcal{V}_1, \dots, \mathcal{V}_m$  are subspaces of  $\mathcal{V}$ . The sum of  $\mathcal{V}_1, \dots, \mathcal{V}_m$ , denoted by  $\mathcal{V}_1 + \dots + \mathcal{V}_m$ , is the set of all possible sums of elements of  $\mathcal{V}_1, \dots, \mathcal{V}_m$ . More precisely,

$$\mathcal{V}_1 + \dots + \mathcal{V}_m = \{v_1 + \dots + v_m : v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m\}$$

**Theorem 1.15.** Suppose  $\mathcal{V}_1, \dots, \mathcal{V}_m$  are subspaces of  $\mathcal{V}$ . Then  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is the smallest subspace of  $\mathcal{V}$  containing  $\mathcal{V}_1, \dots, \mathcal{V}_m$ .

My thoughts

I will add my own proof that sum of subspaces is a subspace (although Axler skips this) because I think it is necessary.

*Proof.* Let us first prove that sum of subspaces is a subspace

- $\mathcal{V}_1 + \dots + \mathcal{V}_m$  contains 0 (additive identity) since all the subspaces contain 0 and sum of subspaces contain every possible sum of their elements.
- Let  $x, y \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$ .  
By definition we can express them as  $x = v_1 + \dots + v_m$  and  $y = u_1 + \dots + u_m$  such that  $u_1, v_1 \in \mathcal{V}_1, \dots, u_m, v_m \in \mathcal{V}_m$ . So we have

$$x + y = (v_1 + \dots + v_m) + (u_1 + \dots + u_m)$$

By commutativity and associativity of vector addition,

$$x + y = (v_1 + u_1) + \dots + (v_m + u_m)$$

Since  $\mathcal{V}_1, \dots, \mathcal{V}_m$  are subspaces,  $(v_1 + u_1) \in \mathcal{V}_1, \dots, (v_m + u_m) \in \mathcal{V}_m$

Hence we conclude  $(x + y) \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, it is closed under addition

- Let  $x \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$ . and  $x = v_1 + \dots + v_m$  where  $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$ . Let  $c$  be any scalar from the field  $\mathbb{F}$ .  
Multiplying  $x$  with the scalar  $c$  yields:

$$cx = c(v_1 + \dots + v_m) = cv_1 + \dots + cv_m$$

Since  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  are subspaces  $cv_1 \in \mathcal{V}_1, \dots, cv_m \in \mathcal{V}_m$ .

We conclude  $cx \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, sum of subspaces is a subspace.

Now we can start with the proof for  $\mathcal{V}_1, \dots, \mathcal{V}_m$  being the smallest subspace of  $\mathcal{V}$  containing  $\mathcal{V}_1, \dots, \mathcal{V}_m$ .

The subspaces  $\mathcal{V}_1, \dots, \mathcal{V}_m$  are all contained in  $\mathcal{V}_1 + \dots + \mathcal{V}_m$ , considering sums  $v_1 + \dots + v_m$  where all except one of the  $v_k$ 's are 0.

Conversely, every subspace of  $\mathcal{V}$  containing  $\mathcal{V}_1, \dots, \mathcal{V}_m$  contains  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  (because subspaces must contain all finite sums of their elements). Thus  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is the smallest subspace of  $\mathcal{V}$  containing  $\mathcal{V}_1, \dots, \mathcal{V}_m$ .  $\square$

### 1.3.2 Direct Sums

**Definition 1.16** (Direct Sum). Suppose  $\mathcal{V}_1, \dots, \mathcal{V}_m$  are subspaces of  $\mathcal{V}$ .

- The sum  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is called a *direct sum* if each element of  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  can be written in only one way as a sum  $v_1 + \dots + v_m$ , where each  $v_k \in \mathcal{V}_k$ .
- If  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is a direct sum, then  $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_m$  denotes  $\mathcal{V}_1 + \dots + \mathcal{V}_m$ , with the  $\oplus$  notation serving as an indication that this is a direct sum.

**Theorem 1.17.** Suppose  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  are subspaces of  $\mathcal{V}$ . Then  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is a direct sum if and only if the only way to write 0 as a sum  $v_1 + \dots + v_m$ , where each  $v_k \in \mathcal{V}_k$ , is by taking each  $v_k$  equal to 0.

*Proof.* First suppose  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum  $v_1 + \dots + v_m$ , where each  $v_k \in \mathcal{V}_k$ , is by taking each  $v_k$  equal to 0.

My thoughts

This is because taking every  $v_k = 0$  is a trivial solution, but since we know it is a direct sum, the only solution would be taking every  $v_k = 0$ .

Now suppose that the only way to write 0 as a sum  $v_1 + \dots + v_m$ , where each  $v_k \in \mathcal{V}_k$ , is by taking each  $v_k$  equal to 0. To show that  $\mathcal{V}_1 + \dots + \mathcal{V}_m$  is a direct sum, let  $v \in \mathcal{V}_1 + \dots + \mathcal{V}_m$ . We can write

$$v = v_1 + \dots + v_m$$

for some  $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$ . To show that this representation is unique, suppose we also have

$$v = u_1 + \dots + u_m$$

where  $u_1 \in \mathcal{V}_1, \dots, u_m \in \mathcal{V}_m$ . Subtracting these two equations, we have

$$0 = (v_1 - u_1) + \dots + (v_m - u_m)$$

Because  $v_1 - u_1 \in \mathcal{V}_1, \dots, v_m - u_m \in \mathcal{V}_m$ , the equation above implies that each  $v_k - u_k$  equals 0. Thus  $v_1 = u_1, \dots, v_m = u_m$ , as desired.  $\square$

**Theorem 1.18.** Suppose  $U$  and  $W$  are subspaces of  $V$ . Then

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

*Proof.* First suppose that  $U + W$  is a direct sum. If  $v \in U \cap W$ , then  $0 = v + (-v)$ , where  $v \in U$  and  $-v \in W$ . By the unique representation of 0 as the sum of a vector in  $U$  and a vector in  $W$ , we have  $v = 0$ . Thus  $U \cap W = \{0\}$ , completing the proof in one direction. To prove the other direction, now suppose  $U \cap W = \{0\}$ . To prove that  $U + W$  is a direct sum, suppose  $u \in U, w \in W$ , and  $0 = u + w$ . To complete the proof, we only need to show that  $u = w = 0$ . The equation above implies that  $u = -w \in W$ . Thus  $u \in U \cap W$ . Hence  $u = 0$ , which by the equation above implies that  $w = 0$ , completing the proof.  $\square$

### 1.4 Exercise 1A

My thoughts

I can't find a way to prove these identities by themselves, maybe I need to learn analysis for that. I will just be assuming  $\mathbb{R}$  is already known to be a field, and use those properties to prove these.

**Question 1.** Show that  $\alpha + \beta = \beta + \alpha$  for all  $\alpha, \beta \in \mathbb{C}$ .

*Proof.* Letting  $\alpha = a + bi$ ,  $\beta = c + di$ ,  $\alpha + \beta = a + ib + c + id$

$$\begin{aligned} &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + id) + (a + ib) \\ &= \beta + \alpha \end{aligned}$$

□

**Question 2.** Show that  $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$  for all  $\alpha, \beta, \lambda \in \mathbb{C}$ .

*Proof.* Letting  $\alpha = a + bi$ ,  $\beta = c + di$ , and  $\lambda = e + fi$  (where  $a, b, c, d, e, f \in \mathbb{R}$ ), the left side of the equation expands as:

$$\begin{aligned} &(\alpha + \beta) + \lambda \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= ((a + c) + e) + ((b + d) + f)i \end{aligned}$$

Because addition is associative in the real numbers, we can rewrite the real and imaginary components as  $(a + (c + e))$  and  $(b + (d + f))$ .

Expanding the right side gives:

$$\begin{aligned} &\alpha + (\beta + \lambda) \\ &= (a + bi) + ((c + e) + (d + f)i) \\ &= (a + (c + e)) + (b + (d + f))i \end{aligned}$$

□

**Question 3.** Show that  $(\alpha\beta)\lambda = \alpha(\beta\lambda)$  for all  $\alpha, \beta, \lambda \in \mathbb{C}$ .

*Proof.* Letting  $\alpha = a + bi$ ,  $\beta = c + di$ , and  $\lambda = e + fi$  (where  $a, b, c, d, e, f \in \mathbb{R}$ ), the left side of the equation expands as:

$$((ac - bd) + (ad + bc)i)(e + fi)$$

Multiplying this out yields a real part of  $ace - bde - adf - bcf$  and an imaginary part of  $acf - bdf + ade + bce$ .

Expanding the right side  $\alpha(\beta\lambda)$  gives

$$(a + bi)((ce - df) + (cf + de)i)$$

Multiplying this out yields a real part of  $ace - adf - bcf - bde$  and an imaginary part of  $acf + ade + bce - bdf$ . □

**Question 4.** Show that  $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$  for all  $\lambda, \alpha, \beta \in \mathbb{C}$

*Proof.* Expanding  $\lambda(\alpha + \beta)$  with  $\lambda = e + fi$ ,  $\alpha = a + bi$ , and  $\beta = c + di$  gives a real part of  $ea + ec - fb - fd$  and an imaginary part of  $eb + ed + fa + fc$ . Multiplying out  $\lambda\alpha + \lambda\beta$  yields the exact same terms, relying on the distributive property of real numbers to match them up. □

**Question 5.** Show that for every  $\alpha \in \mathbb{C}$ , there exists a unique  $\beta \in \mathbb{C}$  such that  $\alpha + \beta = 0$ .

*Proof.* Setting  $\alpha + \beta = 0$  with  $\alpha = a + bi$  and  $\beta = c + di$  results in  $(a + c) + (b + d)i = 0$ . Equating the real and imaginary components to zero means  $a + c = 0$  and  $b + d = 0$ , yielding exactly  $c = -a$  and  $d = -b$ . Because every real number has a uniquely defined additive inverse in  $\mathbb{R}$ , the components  $c$  and  $d$  exist and are unique, making  $\beta$  unique. □

**Question 6.** Show that for every  $\alpha \in \mathbb{C}$  with  $\alpha \neq 0$ , there exists a unique  $\beta \in \mathbb{C}$  such that  $\alpha\beta = 1$ .

*Proof.*  $\alpha = a + bi$ ,  $\beta = c + di$

Setting  $(a + bi)(c + di) = 1$  results in  $(ac - bd) + (ad + bc)i = 1 + 0i$ . Equating the real and imaginary parts gives a system of two linear equations:  $ac - bd = 1$  and  $ad + bc = 0$ . Since  $\alpha \neq 0$ , we know that  $a$  and  $b$  are not both zero, meaning  $a^2 + b^2 \neq 0$ . Solving this system for  $c$  and  $d$  yields exactly  $c = \frac{a}{a^2+b^2}$  and  $d = \frac{-b}{a^2+b^2}$ . This produces a single, well-defined real value for both  $c$  and  $d$ .  $\square$

**Question 7.** Show that  $\frac{-1+\sqrt{3}i}{2}$  is a cube root of 1 (meaning that its cube equals 1).

*Proof.* Squaring  $\frac{-1+\sqrt{3}i}{2}$  yields  $\frac{-1-\sqrt{3}i}{2}$ . Multiplying that squared result by the original term  $\frac{-1+\sqrt{3}i}{2}$  gives  $\frac{(-1)^2 - (\sqrt{3}i)^2}{4} = \frac{1 - (-3)}{4}$ , which reduces perfectly to 1.  $\square$

**Question 8.** Find two distinct square roots of  $i$ .

*Proof.*  $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$  and  $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$   $\square$

**Question 9.** Find  $x \in \mathbb{R}^4$  such that  $(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$ .

*Proof.*  $(1/2, 6, -7/2, 1/2)$   $\square$

**Question 10.** Explain why there does not exist  $\lambda \in \mathbb{C}$  such that  $\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i)$ .

*Proof.* Setting  $\lambda(2 - 3i) = 12 - 5i$  for the first component gives  $\lambda = \frac{12-5i}{2-3i}$ .

Multiplying the numerator and denominator by the conjugate  $(2 + 3i)$  yields  $\frac{24+36i-10i+15}{13}$ , which simplifies to  $\lambda = 3 + 2i$ .

Testing this  $\lambda$  on the second component yields  $(3 + 2i)(5 + 4i) = 15 + 12i + 10i - 8 = 7 + 22i$ , which matches the required vector perfectly.

Now, applying  $\lambda = 3 + 2i$  to the third component gives  $(3 + 2i)(-6 + 7i) = -18 + 21i - 12i - 14 = -32 + 9i$ . But the required third component in the target vector is  $-32 - 9i$ .  $\square$

**Question 11.** Show that  $(x + y) + z = x + (y + z)$  for all  $x, y, z \in \mathbb{F}^n$ .

*Proof.* Let  $x = (x_1, \dots, x_n)$ ,  $y = (y_1, \dots, y_n)$ , and  $z = (z_1, \dots, z_n)$ . Adding component-wise, the left side gives  $((x_1 + y_1) + z_1, \dots, (x_n + y_n) + z_n)$ . Because addition is associative in the underlying field  $\mathbb{F}$ , each component  $(x_i + y_i) + z_i$  is equal to  $x_i + (y_i + z_i)$ . This perfectly constructs the right side  $x + (y + z)$ .  $\square$

**Question 12.** Show that  $(ab)x = a(bx)$  for all  $x \in \mathbb{F}^n$  and all  $a, b \in \mathbb{F}$ .

*Proof.* Expanding  $x$  as  $(x_1, \dots, x_n)$  means the left side  $(ab)x$  becomes  $((ab)x_1, \dots, (ab)x_n)$ . The right side  $a(bx)$  evaluates to  $a(bx_1, \dots, bx_n)$ , which then scales to  $(a(bx_1), \dots, a(bx_n))$ . Because multiplication within the underlying field  $\mathbb{F}$  is associative,  $(ab)x_i = a(bx_i)$  for every component  $\square$

**Question 13.** Show that  $1x = x$  for all  $x \in \mathbb{F}^n$ .

*Proof.* Expanding  $x$  into its components  $(x_1, \dots, x_n)$  gives  $1x = (1x_1, \dots, 1x_n)$ . Because 1 is the multiplicative identity in the field  $\mathbb{F}$ ,  $1x_i = x_i$  for every individual component. This gives us the original vector  $x$ .  $\square$

**Question 14.** Show that  $\lambda(x + y) = \lambda x + \lambda y$  for all  $\lambda \in \mathbb{F}$  and all  $x, y \in \mathbb{F}^n$ .

*Proof.* Expanding  $\lambda(x + y)$  component-wise gives  $(\lambda(x_1 + y_1), \dots, \lambda(x_n + y_n))$ . Because the underlying field  $\mathbb{F}$  has the distributive property, each component expands to  $\lambda x_i + \lambda y_i$ . This perfectly matches the component-wise expansion of the right side,  $\lambda x + \lambda y = (\lambda x_1 + \lambda y_1, \dots, \lambda x_n + \lambda y_n)$ .  $\square$

**Question 15.** Show that  $(a + b)x = ax + bx$  for all  $a, b \in \mathbb{F}$  and all  $x \in \mathbb{F}^n$ .

*Proof.* Expanding  $(a + b)x$  component-wise yields  $((a + b)x_1, \dots, (a + b)x_n)$ . Applying the distributive property of the underlying field  $\mathbb{F}$  to each component gives  $(ax_1 + bx_1, \dots, ax_n + bx_n)$ . This perfectly matches the right side  $ax + bx$  when you add the scaled vectors  $(ax_1, \dots, ax_n)$  and  $(bx_1, \dots, bx_n)$  together.  $\square$

## 1.5 Exercise 1B

**Question 1.** Prove that  $-(-v) = v$  for every  $v \in \mathcal{V}$ .

*Proof.*

$$\begin{aligned} -(-v) &= -(-v) + 0 \\ -(-v) &= -(-v) + (-v + v) \\ -(-v) &= (-(-v) + -v) + v \\ -(-v) &= 0 + v \\ -(-v) &= v \end{aligned}$$

$\square$

**Question 2.** Suppose  $a \in \mathbb{F}, v \in \mathcal{V}$ , and  $av = 0$ . Prove that  $a = 0$  or  $v = 0$ .

*Proof.* Given equation:

$$av = 0$$

Since  $a \in \mathbb{F}$  (a field) and  $a \neq 0$ , it has a multiplicative inverse  $a^{-1} \in \mathbb{F}$ .

$$\begin{aligned} a^{-1}(av) &= a^{-1}(0) \\ a^{-1}(av) &= 0 \end{aligned}$$

By associativity of scalar multiplication:

$$\begin{aligned} (a^{-1}a)v &= 0 \\ 1v &= 0 \\ v &= 0 \end{aligned}$$

If  $a \neq 0$ , then  $v$  must be 0. Therefore, if  $av = 0$ , it must be true that either  $a = 0$  or  $v = 0$ .  $\square$

**Question 3.** Suppose  $v, w \in \mathcal{V}$ . Explain why there exists a unique  $x \in \mathcal{V}$  such that  $v + 3x = w$ .

*Proof.* First we prove existence:

$$x = \frac{1}{3}(w - v)$$

Since a vector space is closed under addition and scalar multiplication  $x \in \mathcal{V}$

Now for uniqueness:

Suppose there are two vectors,  $x_1$  and  $x_2$  in  $V$ , that both satisfy the equation.

$$\begin{aligned} v + 3x_1 &= v + 3x_2 \\ -v + (v + 3x_1) &= -v + (v + 3x_2) \end{aligned}$$

Using associativity and additive inverse properties,

$$\begin{aligned} 3x_1 &= 3x_2 \\ x_1 &= x_2 \end{aligned}$$

Thus  $x$  is unique.  $\square$

**Question 4.** The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

*Proof.* It doesn't have the zero element.  $\square$

**Question 5.** Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that  $0v = 0$  for all  $v \in \mathcal{V}$ .

*Proof.* Let's assume the additive inverse condition does not exist. Instead we have  $0v = 0$  for all  $v \in \mathcal{V}$ . We can arrive at the additive inverse criterion from here.

$$0v = 0$$

$$(1 + (-1))v = 0$$

Using distributivity,

$$1v + (-1)v = 0$$

Using multiplicative identity,

$$v + (-1)v = 0$$

The element  $(-1)v$  satisfies the exact requirements of an additive inverse.  $\square$

**Question 6.** Let  $\infty$  and  $-\infty$  denote two distinct objects, neither of which is in  $\mathbb{R}$ . Define an addition and scalar multiplication on  $\mathbb{R} \cup \{\infty, -\infty\}$  as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for  $t \in \mathbb{R}$  define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is  $\mathbb{R} \cup \{\infty, -\infty\}$  a vector space over  $\mathbb{R}$ ? Explain.

*Proof.* It is not a vector space.

If  $t + \infty = \infty$ , we have:

$$\begin{aligned} 1 + \infty &= 2 + \infty \\ 1 &= 2 \end{aligned}$$

We can cancel out the  $\infty$  on both sides due to existence of additive inverse.  $\square$

**Question 7.** Suppose  $S$  is a nonempty set. Let  $\mathcal{V}^S$  denote the set of functions from  $S$  to  $\mathcal{V}$ . Define a natural addition and scalar multiplication on  $\mathcal{V}^S$ , and show that  $\mathcal{V}^S$  is a vector space with these definitions.

*Proof.* Because  $\mathcal{V}$  is a vector space,  $\mathcal{V}^S$  will obey all vector space axioms through point-wise evaluations for every  $x \in S$ .

- Commutativity & Associativity:  $(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$ . Associativity follows the exact same logic.

- Additive Identity: The zero function, defined by  $0(x) = 0$  for all  $x \in S$ .
- Additive Inverse: The function defined point-wise by  $(-f)(x) = -(f(x))$ .
- Distributive Properties:  $a(f + g)(x) = a(f(x) + g(x)) = af(x) + ag(x) = (af + ag)(x)$ .

□

**Question 8.** Suppose  $\mathcal{V}$  is a real vector space.

- The *complexification* of  $\mathcal{V}$ , denoted by  $\mathcal{V}_{\mathbb{C}}$ , equals  $\mathcal{V} \times \mathcal{V}$ . An element of  $\mathcal{V}_{\mathbb{C}}$  is an ordered pair  $(u, v)$ , where  $u, v \in \mathcal{V}$ , but we write this as  $u + iv$ .
- Addition on  $\mathcal{V}_{\mathbb{C}}$  is defined by

$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$

for all  $u_1, v_1, u_2, v_2 \in \mathcal{V}$ .

- Complex scalar multiplication on  $\mathcal{V}_{\mathbb{C}}$  is defined by

$$(a + bi)(u + iv) = (au - bv) + i(av + bu)$$

for all  $a, b \in \mathbb{R}$  and all  $u, v \in \mathcal{V}$ .

Prove that with the definitions of addition and scalar multiplication as above,  $\mathcal{V}_{\mathbb{C}}$  is a complex vector space.

*Proof.* Because  $\mathcal{V}$  is a real vector space, it is inherently closed under its own addition and scalar multiplication. Therefore,  $(u_1 + u_2) \in \mathcal{V}$ ,  $(v_1 + v_2) \in \mathcal{V}$ ,  $(au - bv) \in \mathcal{V}$ , and  $(av + bu) \in \mathcal{V}$ , guaranteeing the operations always produce valid elements of  $\mathcal{V}_{\mathbb{C}}$ .

The following properties are also satisfied:

- Additive Identity:  $0 + i0$  (since  $0 \in \mathcal{V}$ ).
- Additive Inverse:  $-(u) + i(-v)$  (since  $\mathcal{V}$  contains inverses for  $u$  and  $v$ ).
- Multiplicative Identity:  $1 + i0$  acts as the scalar 1, yielding  $1u - 0v + i(1v + 0u) = u + iv$ .
- Associativity, Commutativity, and Distributivity: These expand algebraically and hold true because the vectors in  $\mathcal{V}$  and scalars in  $\mathbb{R}$  already obey these rules.

□

## 1.6 Exercise 1C

**Question 1.** For each of the following subsets of  $\mathbf{F}^3$ , determine whether it is a subspace of  $\mathbf{F}^3$ .

- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

*Proof.* We only need to check for closure under addition and scalar multiplication and existence of a zero element. By this logic, (a),(d) are subspaces while (b),(c) are not. □

**Question 2.** Verify all assertions about subspaces below

- (a) If  $b \in \mathbb{F}$ , then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of  $\mathbb{F}^4$  if and only if  $b = 0$ .

- (b) The set of continuous real-valued functions on the interval  $[0, 1]$  is a subspace of  $\mathbb{R}^{[0,1]}$ .  
 (c) The set of differentiable real-valued functions on  $\mathbb{R}$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$ .  
 (d) The set of differentiable real-valued functions  $f$  on the interval  $(0, 3)$  such that  $f'(2) = b$  is a subspace of  $\mathbb{R}^{(0,3)}$  if and only if  $b = 0$ .  
 (e) The set of all sequences of complex numbers with limit 0 is a subspace of  $\mathbb{C}^{\infty}$ .

*Proof.* (a) Forward direction (If it is a subspace, then  $b = 0$ ): Every subspace must contain the additive identity, which in  $\mathbb{F}^4$  is the zero vector  $(0, 0, 0, 0)$ . Substituting these coordinates into the defining equation yields  $0 = 5(0) + b$ , which simplifies strictly to  $b = 0$ .

Reverse direction (If  $b = 0$ , then it is a subspace): Setting  $b = 0$  changes the subset condition to  $x_3 = 5x_4$ . Clearly, this is closed under addition, scalar multiplication and the zero vector  $(0, 0, 0, 0)$  exists.

- (b) To prove that  $W$  is a subspace of  $\mathbb{R}^{[0,1]}$ , we must verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in  $\mathbb{R}^{[0,1]}$  is the zero function, denoted by  $\mathbf{0}$ , where  $\mathbf{0}(x) = 0$  for all  $x \in [0, 1]$ . Because constant functions are continuous on  $\mathbb{R}$ , the zero function is continuous on  $[0, 1]$ . Therefore,  $\mathbf{0} \in W$ , proving that  $W$  is non-empty.  
 (b) Closure under vector addition: Let  $f, g \in W$ . By definition,  $f$  and  $g$  are both continuous functions on  $[0, 1]$ . From the properties of continuous functions in real analysis, the sum of two continuous functions is continuous. Therefore, the function  $(f + g)(x) = f(x) + g(x)$  is continuous on  $[0, 1]$ , which implies  $f + g \in W$ .  
 (c) Closure under scalar multiplication: Let  $f \in W$  and  $c \in \mathbb{R}$ . Since  $f$  is a continuous function on  $[0, 1]$ , its multiplication by a scalar constant  $c$  results in a continuous function. Thus, the function  $(cf)(x) = cf(x)$  is continuous on  $[0, 1]$ , meaning  $cf \in W$ .

Since  $W$  contains the zero vector and is closed under both addition and scalar multiplication,  $W$  is a subspace of  $\mathbb{R}^{[0,1]}$ .

- (c) To prove that  $W$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$ , we verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in  $\mathbb{R}^{\mathbb{R}}$  is the zero function, denoted by  $\mathbf{0}$ , where  $\mathbf{0}(x) = 0$  for all  $x \in \mathbb{R}$ . Because constant functions are differentiable everywhere on  $\mathbb{R}$  (with a derivative of 0), the zero function is differentiable. Therefore,  $\mathbf{0} \in W$ , proving that  $W$  is non-empty.  
 (b) Closure under vector addition: Let  $f, g \in W$ . By definition,  $f$  and  $g$  are both differentiable functions on  $\mathbb{R}$ . From standard calculus theorems, the sum of two differentiable functions is differentiable, and its derivative is given by  $(f + g)'(x) = f'(x) + g'(x)$ . Therefore, the function  $(f + g)$  is differentiable on  $\mathbb{R}$ , which implies  $f + g \in W$ .  
 (c) Closure under scalar multiplication: Let  $f \in W$  and  $c \in \mathbb{R}$ . Since  $f$  is a differentiable function on  $\mathbb{R}$ , its multiplication by a scalar constant  $c$  results in a differentiable function, with the derivative  $(cf)'(x) = cf'(x)$ . Thus, the function  $cf$  is differentiable on  $\mathbb{R}$ , meaning  $cf \in W$ .

Since  $W$  contains the zero vector and is closed under both vector addition and scalar multiplication,  $W$  is a subspace of  $\mathbb{R}^{\mathbb{R}}$ .

(d) Let  $\mathcal{V} = \mathbb{R}^{(0,3)}$  and let  $W_b = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = b\}$ . We must prove that  $W_b$  is a subspace of  $\mathcal{V}$  if and only if  $b = 0$ .

( $\Rightarrow$ ) Assume  $W_b$  is a subspace of  $\mathcal{V}$ . Then it must contain the zero vector of  $\mathcal{V}$ , which is the zero function  $\mathbf{0}$  defined by  $\mathbf{0}(x) = 0$  for all  $x \in (0,3)$ . The derivative of the zero function is  $\mathbf{0}'(x) = 0$  for all  $x \in (0,3)$ . Since  $\mathbf{0} \in W_b$ , it must satisfy the condition defining the set  $W_b$ , which means  $\mathbf{0}'(2) = b$ . Thus,  $0 = b$ .

( $\Leftarrow$ ) Assume  $b = 0$ . We must show that  $W_0 = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = 0\}$  is a subspace. We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero function  $\mathbf{0}$  is differentiable, and as shown above,  $\mathbf{0}'(2) = 0$ . Thus,  $\mathbf{0} \in W_0$ .
2. Closure under vector addition: Let  $f, g \in W_0$ . Both are differentiable on  $(0,3)$ , so their sum  $f + g$  is also differentiable. By the linearity of the derivative,  $(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0$ . Therefore,  $f + g \in W_0$ .
3. Closure under scalar multiplication: Let  $f \in W_0$  and  $c \in \mathbb{R}$ . The function  $cf$  is differentiable, and its derivative at  $x = 2$  is  $(cf)'(2) = cf'(2) = c(0) = 0$ . Therefore,  $cf \in W_0$ .

Since  $W_0$  contains the zero vector and is closed under both addition and scalar multiplication, it is a subspace of  $\mathcal{V}$ .

Therefore,  $W_b$  is a subspace of  $\mathbb{R}^{(0,3)}$  if and only if  $b = 0$ .

#### My thoughts

I don't know anything about complex sequences, but it is easy enough to wager a guess after a google search.

(e)

Let  $\mathbb{C}^\infty$  be the vector space of all sequences of complex numbers, and let  $W$  be the set of all sequences of complex numbers with limit 0 [cite: 3]. We will prove that  $W$  is a subspace of  $\mathbb{C}^\infty$ .

We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero vector in  $\mathbb{C}^\infty$  is the zero sequence, denoted by  $\mathbf{0} = (0, 0, 0, \dots)$ , where  $z_n = 0$  for all  $n \geq 1$ . The limit of this constant sequence is  $\lim_{n \rightarrow \infty} 0 = 0$ . Therefore,  $\mathbf{0} \in W$ , meaning the subset is non-empty.
2. Closure under vector addition: Let  $x = (x_n)$  and  $y = (y_n)$  be sequences in  $W$ . By definition,  $\lim_{n \rightarrow \infty} x_n = 0$  and  $\lim_{n \rightarrow \infty} y_n = 0$ . According to the algebraic limit theorems for sequences, the limit of their sum is given by  $\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = 0 + 0 = 0$ . Thus, the sequence  $x + y \in W$ .
3. Closure under scalar multiplication: Let  $x = (x_n) \in W$  and let  $\lambda \in \mathbb{C}$ . By definition,  $\lim_{n \rightarrow \infty} x_n = 0$ . Using the limit laws for scalar multiples,  $\lim_{n \rightarrow \infty} (\lambda x_n) = \lambda \lim_{n \rightarrow \infty} x_n = \lambda \cdot 0 = 0$ . Therefore, the sequence  $\lambda x \in W$ .

So  $W$  is a subspace of  $\mathbb{C}^\infty$ . □

**Question 3.** Show that the set of differentiable real-valued functions  $f$  on the interval  $(-4, 4)$  such that  $f'(-1) = 3f(2)$  is a subspace of  $\mathbf{R}^{(-4,4)}$ .

*Proof.* Let  $W = \{f \in V \mid f \text{ is differentiable on } (-4, 4) \text{ and } f'(-1) = 3f(2)\}$ . We verify that  $W$  satisfies the three conditions of a subspace:

1. **Additive identity:** The zero function, denoted by  $\mathbf{0}$ , where  $\mathbf{0}(x) = 0$  for all  $x \in (-4, 4)$ , is differentiable. Its derivative is everywhere zero, so  $\mathbf{0}'(-1) = 0$ . Evaluating the other side gives  $3\mathbf{0}(2) = 3(0) = 0$ . Thus,  $\mathbf{0}'(-1) = 3\mathbf{0}(2)$ , meaning  $\mathbf{0} \in W$ .
2. **Closed under addition:** Let  $f, g \in W$ . Since  $f$  and  $g$  are differentiable, their sum  $f + g$  is also differentiable. By the linearity of the derivative, we have:

$$(f + g)'(-1) = f'(-1) + g'(-1)$$

Because  $f, g \in W$ , we can substitute  $f'(-1) = 3f(2)$  and  $g'(-1) = 3g(2)$ :

$$(f + g)'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2)$$

Therefore,  $f + g \in W$ .

3. **Closed under scalar multiplication:** Let  $f \in W$  and  $\lambda \in \mathbb{R}$ . The scalar multiple  $\lambda f$  is differentiable, and by the linearity of the derivative:

$$(\lambda f)'(-1) = \lambda f'(-1)$$

Since  $f \in W$ , we know  $f'(-1) = 3f(2)$ , yielding:

$$(\lambda f)'(-1) = \lambda(3f(2)) = 3(\lambda f(2)) = 3(\lambda f)(2)$$

Therefore,  $\lambda f \in W$ .

Since  $W$  contains the zero vector and is closed under both addition and scalar multiplication,  $W$  is a subspace of  $\mathbb{R}^{(-4,4)}$ .  $\square$

**Question 4.** Suppose  $b \in \mathbb{R}$ . Show that the set of continuous real-valued functions  $f$  on the interval  $[0, 1]$  such that  $\int_0^1 f = b$  is a subspace of  $\mathbb{R}^{[0,1]}$  if and only if  $b = 0$ .

*Proof.* Let  $W_b = \left\{ f \in V \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = b \right\}$ . We want to show that  $W_b$  is a subspace of  $V$  if and only if  $b = 0$ .

( $\Rightarrow$ ) Assume  $W_b$  is a subspace of  $\mathbb{R}^{[0,1]}$ . Then  $W_b$  must contain the additive identity of  $\mathbb{R}^{[0,1]}$ , which is the zero function  $\mathbf{0}$  defined by  $\mathbf{0}(x) = 0$  for all  $x \in [0, 1]$ . The zero function is continuous, and its integral is:

$$\int_0^1 \mathbf{0} = \int_0^1 0 \, dx = 0$$

Since  $\mathbf{0} \in W_b$ , it must satisfy the condition defining the set, meaning  $\int_0^1 \mathbf{0} = b$ . Therefore,  $b = 0$ .

( $\Leftarrow$ ) Assume  $b = 0$ . We must show that  $W_0 = \left\{ f \in \mathbb{R}^{[0,1]} \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = 0 \right\}$  is a subspace. We verify the three conditions:

1. **Additive identity:** As shown above, the zero function  $\mathbf{0}$  is continuous and  $\int_0^1 \mathbf{0} = 0$ . Thus,  $\mathbf{0} \in W_0$ .
2. **Closed under addition:** Let  $f, g \in W_0$ . Since both are continuous, their sum  $f + g$  is also continuous. By the linearity of the definite integral:

$$\int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0$$

Therefore,  $f + g \in W_0$ .

3. **Closed under scalar multiplication:** Let  $f \in W_0$  and  $\lambda \in \mathbb{R}$ . The scalar multiple  $\lambda f$  is continuous, and by the linearity of the integral:

$$\int_0^1 (\lambda f) = \lambda \int_0^1 f = \lambda(0) = 0$$

Therefore,  $\lambda f \in W_0$ .

Thus,  $W_b$  is a subspace of  $\mathbb{R}^{[0,1]}$  if and only if  $b = 0$ . □

**Question 5.** Is  $\mathbb{R}^2$  a subspace of the complex vector space  $\mathbb{C}^2$ ?

*Proof.* No,  $\mathbb{R}^2$  is not a subspace of the complex vector space  $\mathbb{C}^2$ .

To be a subspace of the complex vector space  $\mathbb{C}^2$ , the subset  $\mathbb{R}^2$  must be closed under scalar multiplication for all scalars in the field  $\mathbb{C}$ .

Consider the vector  $v = (1, 0) \in \mathbb{R}^2$  and the complex scalar  $\lambda = i \in \mathbb{C}$ . Multiplying the vector by the scalar yields:

$$\lambda v = i(1, 0) = (i, 0)$$

Since  $i \notin \mathbb{R}$ , the resulting vector  $(i, 0)$  is not in  $\mathbb{R}^2$ . □

**Question 6.** (a) Is  $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$  a subspace of  $\mathbb{R}^3$ ?  
 (b) Is  $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$  a subspace of  $\mathbb{C}^3$ ?

*Proof.* **(a) Yes, it is a subspace.** Let  $W = \{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ . For real numbers, the function  $f(x) = x^3$  is injective, so  $a^3 = b^3$  implies  $a = b$ . Thus,  $W$  can be rewritten as  $W = \{(a, a, c) \mid a, c \in \mathbb{R}\}$ . We verify the subspace conditions:

1. Additive identity: The zero vector  $(0, 0, 0)$  is in  $W$  since  $0 = 0$ .
2. Closed under addition: Let  $v_1 = (a_1, a_1, c_1)$  and  $v_2 = (a_2, a_2, c_2)$  be in  $W$ . Their sum  $v_1 + v_2 = (a_1 + a_2, a_1 + a_2, c_1 + c_2)$  is in  $W$  because its first two coordinates are equal.
3. Closed under scalar multiplication: Let  $v = (a, a, c) \in W$  and  $\lambda \in \mathbb{R}$ . The scalar multiple  $\lambda v = (\lambda a, \lambda a, \lambda c)$  is in  $W$  because its first two coordinates are equal.

Therefore,  $W$  is a subspace of  $\mathbb{R}^3$ .

**(b) No, it is not a subspace.** Consider the vectors  $v_1 = (1, 1, 0)$  and  $v_2 = (\omega, 1, 0)$ . Both vectors are in  $U$  because  $1^3 = 1^3$  and  $\omega^3 = 1 = 1^3$ . Their sum is:

$$v_1 + v_2 = (1 + \omega, 2, 0)$$

For this sum to be in  $U$ , its coordinates must satisfy  $a^3 = b^3$ , meaning we must have  $(1 + \omega)^3 = 2^3 = 8$ . Expanding the left side:

$$(1 + \omega)^3 = 1^3 + 3(1^2)\omega + 3(1)\omega^2 + \omega^3 = 1 + 3\omega + 3\omega^2 + 1 = 2 + 3(\omega + \omega^2)$$

Since  $1 + \omega + \omega^2 = 0$  for cube roots of unity, we know  $\omega + \omega^2 = -1$ . Thus:

$$(1 + \omega)^3 = 2 + 3(-1) = -1$$

Since  $-1 \neq 8$ , the vector  $v_1 + v_2 \notin U$ . Because  $U$  is not closed under addition, it is not a subspace of  $\mathbb{C}^3$ . □

**Question 7.** Prove or give a counterexample: If  $U$  is a nonempty subset of  $\mathbb{R}^2$  such that  $U$  is closed under addition and under taking additive inverses (meaning  $-u \in U$  whenever  $u \in U$ ), then  $U$  is a subspace of  $\mathbb{R}^2$ .

*Proof.* The statement is false. We provide a counterexample.

Let  $U = \mathbb{Z}^2 = \{(m, n) \mid m, n \in \mathbb{Z}\}$ , which is the set of all vectors in  $\mathbb{R}^2$  with integer coordinates.

We verify that  $U$  satisfies the given conditions:

1.  **$U$  is nonempty:** The vector  $(0, 0) \in U$  because 0 is an integer.
2. **Closed under addition:** Let  $u = (m_1, n_1) \in U$  and  $v = (m_2, n_2) \in U$ . Their sum is  $u + v = (m_1 + m_2, n_1 + n_2)$ . Since the sum of two integers is an integer,  $u + v \in U$ .
3. **Closed under additive inverses:** Let  $u = (m, n) \in U$ . Its additive inverse is  $-u = (-m, -n)$ . Since the negative of an integer is an integer,  $-u \in U$ .

However,  $U$  is not a subspace of  $\mathbb{R}^2$  because it is not closed under scalar multiplication over the field  $\mathbb{R}$ .

For a specific counterexample, let  $u = (1, 1) \in U$  and let the scalar  $\lambda = \frac{1}{2} \in \mathbb{R}$ . Multiplying them yields:

$$\lambda u = \frac{1}{2}(1, 1) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Because  $\frac{1}{2}$  is not an integer, the resulting vector  $\left(\frac{1}{2}, \frac{1}{2}\right) \notin U$ .

Since  $U$  is not closed under scalar multiplication, it fails the subspace criterion. Therefore, the statement is false.  $\square$

**Question 8.** Give an example of a nonempty subset  $U$  of  $\mathbb{R}^2$  such that  $U$  is closed under scalar multiplication, but  $U$  is not a subspace of  $\mathbb{R}^2$ .

*Proof.*  $U = \{(x, y) \in \mathbb{R}^2 \mid x = 0 \text{ or } y = 0\}$   $\square$

**Question 9.** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called *periodic* if there exists a positive number  $p$  such that  $f(x) = f(x + p)$  for all  $x \in \mathbb{R}$ . Is the set of periodic functions from  $\mathbb{R}$  to  $\mathbb{R}$  a subspace of  $\mathbb{R}^{\mathbb{R}}$ ? Explain.

*Proof.* We provide a counterexample. Let  $f(x) = \cos(x)$  and  $g(x) = \cos(\pi x)$ . Both  $f$  and  $g$  are periodic functions:

- $f$  has a period of  $2\pi$ , since  $\cos(x + 2\pi) = \cos(x)$ .
- $g$  has a period of 2, since  $\cos(\pi(x + 2)) = \cos(\pi x + 2\pi) = \cos(\pi x)$ .

Let  $h = f + g$ , so  $h(x) = \cos(x) + \cos(\pi x)$ . We will show that  $h$  is not periodic. Suppose, for the sake of contradiction, that  $h$  is periodic with some positive period  $p > 0$ . Then  $h(x) = h(x + p)$  for all  $x \in \mathbb{R}$ . Evaluating at  $x = 0$ , we get:

$$h(p) = h(0)$$

$$\cos(p) + \cos(\pi p) = \cos(0) + \cos(\pi \cdot 0) = 2$$

Since the maximum value of the real-valued cosine function is 1, the sum  $\cos(p) + \cos(\pi p)$  can only equal 2 if both terms are exactly 1:

$$\cos(p) = 1 \quad \text{and} \quad \cos(\pi p) = 1$$

The condition  $\cos(p) = 1$  implies that  $p = 2\pi k$  for some positive integer  $k$ . The condition  $\cos(\pi p) = 1$  implies that  $\pi p = 2\pi m$ , which simplifies to  $p = 2m$  for some positive integer  $m$ .

Equating the two expressions for  $p$ , we have:

$$2\pi k = 2m \implies \pi = \frac{m}{k}$$

Because  $k$  and  $m$  are positive integers, this implies that  $\pi$  is a rational number. This is a contradiction, as  $\pi$  is known to be irrational.

Therefore, our assumption that  $h$  is periodic must be false. Because  $f$  and  $g$  are periodic but their sum  $f + g$  is not, the set of periodic functions is not closed under addition. Consequently, it is not a subspace of  $\mathbb{R}^{\mathbb{R}}$ .  $\square$

**Question 10.** Suppose  $\mathcal{V}_1$  and  $\mathcal{V}_2$  are subspaces of  $\mathcal{V}$ . Prove that the intersection  $\mathcal{V}_1 \cap \mathcal{V}_2$  is a subspace of  $\mathcal{V}$ .

*Proof.* Let  $\mathcal{V}_1$  and  $\mathcal{V}_2$  be subspaces of a vector space  $\mathcal{V}$  over a field  $\mathbb{F}$ .

1. **Additive identity:** Because  $\mathcal{V}_1$  and  $\mathcal{V}_2$  are subspaces of  $\mathcal{V}$ , both must contain the additive identity of  $\mathcal{V}$ . Thus,  $0 \in \mathcal{V}_1$  and  $0 \in \mathcal{V}_2$ . Since  $0$  is in both sets, it is in their intersection, meaning  $0 \in \mathcal{V}_1 \cap \mathcal{V}_2$ .
2. **Closed under addition:** Let  $u, w \in \mathcal{V}_1 \cap \mathcal{V}_2$ . By definition of the intersection,  $u, w \in \mathcal{V}_1$  and  $u, w \in \mathcal{V}_2$ . Because  $\mathcal{V}_1$  is a subspace, it is closed under addition, so  $u + w \in \mathcal{V}_1$ . Similarly, because  $\mathcal{V}_2$  is a subspace,  $u + w \in \mathcal{V}_2$ . Since  $u + w$  is in both  $\mathcal{V}_1$  and  $\mathcal{V}_2$ , it follows that  $u + w \in \mathcal{V}_1 \cap \mathcal{V}_2$ .
3. **Closed under scalar multiplication:** Let  $u \in \mathcal{V}_1 \cap \mathcal{V}_2$  and let  $\lambda \in \mathbb{F}$ . By definition of the intersection,  $u \in \mathcal{V}_1$  and  $u \in \mathcal{V}_2$ . Because  $\mathcal{V}_1$  and  $\mathcal{V}_2$  are subspaces, both are closed under scalar multiplication. Therefore,  $\lambda u \in \mathcal{V}_1$  and  $\lambda u \in \mathcal{V}_2$ . Since  $\lambda u$  is in both sets,  $\lambda u \in \mathcal{V}_1 \cap \mathcal{V}_2$ .

$\square$

**Question 11.** Prove that the intersection of every collection of subspaces of  $V$  is a subspace of  $V$ .

*Proof.* Let  $\{V_i\}_{i \in I}$  be a collection of subspaces of a vector space  $V$  over a field  $\mathbb{F}$ , where  $I$  is an index set. Let  $W = \bigcap_{i \in I} V_i$  be the intersection of this collection. We verify that  $W$  satisfies the three conditions of a subspace:

1. **Additive identity:** Since each  $V_i$  is a subspace of  $\mathcal{V}$ , the additive identity  $0$  must be in every  $V_i$  for all  $i \in I$ . Because  $0$  is in every subspace in the collection, it is in their intersection. Thus,  $0 \in W$ .
2. **Closed under addition:** Let  $u, w \in W$ . By the definition of the intersection,  $u \in V_i$  and  $w \in V_i$  for every  $i \in I$ . Because every  $V_i$  is a subspace, they are all individually closed under addition, meaning  $u + w \in V_i$  for all  $i \in I$ . Since  $u + w$  is in every subspace in the collection,  $u + w \in \bigcap_{i \in I} V_i = W$ .
3. **Closed under scalar multiplication:** Let  $u \in W$  and  $\lambda \in \mathbb{F}$ . By definition,  $u \in V_i$  for all  $i \in I$ . Because each  $V_i$  is a subspace, it is closed under scalar multiplication, so  $\lambda u \in V_i$  for all  $i \in I$ . Since  $\lambda u$  is in every subspace in the collection,  $\lambda u \in \bigcap_{i \in I} V_i = W$ .

Since the intersection  $W$  contains the additive identity and is closed under both addition and scalar multiplication, it is a subspace of  $\mathcal{V}$ .  $\square$

**Question 12.** Prove that the union of two subspaces of  $\mathcal{V}$  is a subspace of  $\mathcal{V}$  if and only if one of the subspaces is contained in the other.

*Proof.* Let  $V_1$  and  $V_2$  be subspaces of a vector space  $V$ . We must prove that  $V_1 \cup V_2$  is a subspace if and only if  $V_1 \subseteq V_2$  or  $V_2 \subseteq V_1$ .

( $\Leftarrow$ ) Assume one of the subspaces is contained in the other. Without loss of generality, let  $V_1 \subseteq V_2$ . Then their union is  $V_1 \cup V_2 = V_2$ . Since  $V_2$  is given to be a subspace of  $V$ , it follows immediately that  $V_1 \cup V_2$  is a subspace of  $V$ .

( $\Rightarrow$ ) Assume that the union  $V_1 \cup V_2$  is a subspace of  $V$ . We want to show that either  $V_1 \subseteq V_2$  or  $V_2 \subseteq V_1$ .

Suppose, for the sake of contradiction, that neither subspace is contained in the other. This means there exists some vector  $u \in V_1$  such that  $u \notin V_2$ , and there exists some vector  $w \in V_2$  such that  $w \notin V_1$ .

Since  $u \in V_1$  and  $w \in V_2$ , both vectors are in  $V_1 \cup V_2$ . Because we assumed  $V_1 \cup V_2$  is a subspace, it must be closed under vector addition. Therefore, the sum  $u + w$  must be in  $V_1 \cup V_2$ . By the definition of the union, this implies that either  $u + w \in V_1$  or  $u + w \in V_2$ .

**Case 1:** Suppose  $u + w \in V_1$ . Since  $V_1$  is a subspace, it is closed under additive inverses and addition, so it must contain  $-u$ . Therefore,  $(u + w) + (-u) \in V_1$ , which simplifies to  $w \in V_1$ . This directly contradicts our initial assumption that  $w \notin V_1$ .

**Case 2:** Suppose  $u + w \in V_2$ . Similarly, since  $V_2$  is a subspace, it contains  $-w$ . Therefore,  $(u + w) + (-w) \in V_2$ , which simplifies to  $u \in V_2$ . This directly contradicts our initial assumption that  $u \notin V_2$ .

In both cases, we arrive at a contradiction. Therefore, our assumption that neither subspace is contained in the other must be false. We conclude that  $V_1 \subseteq V_2$  or  $V_2 \subseteq V_1$ .  $\square$

**Question 13.** Prove that the union of three subspaces of  $\mathcal{V}$  is a subspace of  $\mathcal{V}$  if and only if one of the subspaces contains the other two.

*Proof.* Let  $W = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3$ . We want to prove that  $W$  is a subspace of  $\mathcal{V}$  if and only if one of the subspaces contains the other two. (Note: This proof assumes the underlying field  $\mathbb{F}$  is  $\mathbb{R}$  or  $\mathbb{C}$ , which means it has more than 2 elements).

( $\Leftarrow$ ) Assume one subspace contains the other two. Without loss of generality, let  $\mathcal{V}_1$  contain  $\mathcal{V}_2$  and  $\mathcal{V}_3$ . Then the union  $\mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3 = \mathcal{V}_1$ . Because  $\mathcal{V}_1$  is given to be a subspace of  $\mathcal{V}$ , their union is trivially a subspace of  $\mathcal{V}$ .

( $\Rightarrow$ ) Assume  $W = \mathcal{V}_1 \cup \mathcal{V}_2 \cup \mathcal{V}_3$  is a subspace of  $\mathcal{V}$ . We want to show that one of the subspaces contains the other two.

Suppose for the sake of contradiction that no single subspace contains the other two. If the union of any two of these subspaces completely covered  $W$  (for example, if  $W = \mathcal{V}_1 \cup \mathcal{V}_2$ ), then by the known theorem for the union of two subspaces, we would have  $\mathcal{V}_1 \subseteq \mathcal{V}_2$  or  $\mathcal{V}_2 \subseteq \mathcal{V}_1$ . This would mean  $W = \mathcal{V}_2$  or  $W = \mathcal{V}_1$ , implying that one subspace contains all three. Since we assumed no single subspace contains the other two, no two subspaces can cover  $W$ .

Therefore, there must exist vectors that strictly belong to only one of the subspaces:

- There is some  $v_1 \in \mathcal{V}_1$  such that  $v_1 \notin \mathcal{V}_2 \cup \mathcal{V}_3$ .
- There is some  $v_2 \in \mathcal{V}_2$  such that  $v_2 \notin \mathcal{V}_1 \cup \mathcal{V}_3$ .

Because  $W$  is a subspace, it is closed under vector addition and scalar multiplication. Let  $c$  be any non-zero scalar. The vector  $v_1 + cv_2$  must be in  $W$ , meaning it is in  $\mathcal{V}_1$ ,  $\mathcal{V}_2$ , or  $\mathcal{V}_3$ .

- If  $v_1 + cv_2 \in \mathcal{V}_1$ : Since  $\mathcal{V}_1$  is a subspace, it contains  $-v_1$ . Thus,  $(v_1 + cv_2) - v_1 = cv_2 \in \mathcal{V}_1$ . Because  $c \neq 0$ , this implies  $v_2 \in \mathcal{V}_1$ , which contradicts our choice of  $v_2$ .
- If  $v_1 + cv_2 \in \mathcal{V}_2$ : Since  $\mathcal{V}_2$  is a subspace, it contains  $-cv_2$ . Thus,  $(v_1 + cv_2) - cv_2 = v_1 \in \mathcal{V}_2$ , which contradicts our choice of  $v_1$ .

Thus, for every non-zero scalar  $c$ , the vector  $v_1 + cv_2$  must belong to  $\mathcal{V}_3$ .

Because our field  $\mathbb{F}$  is  $\mathbb{R}$  or  $\mathbb{C}$ , there are infinitely many scalars. We can pick two distinct non-zero scalars, say  $c = 1$  and  $c = 2$ . Based on the logic above, both  $v_1 + v_2 \in \mathcal{V}_3$  and  $v_1 + 2v_2 \in \mathcal{V}_3$ . Since  $\mathcal{V}_3$  is a subspace, it is closed under vector subtraction, meaning:

$$(v_1 + 2v_2) - (v_1 + v_2) = v_2 \in \mathcal{V}_3$$

But this directly contradicts our initial choice that  $v_2 \notin \mathcal{V}_1 \cup \mathcal{V}_3$ .

This contradiction means our assumption that no subspace contains the other two must be false. Therefore, one of the subspaces must contain the other two.  $\square$

**Question 14.** Suppose

$$U = \{(x, -x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbb{F}^3 : x \in \mathbb{F}\}.$$

Describe  $U + W$  using symbols, and also give a description of  $U + W$  that uses no symbols.

*Proof.* Let  $u = (x, -x, 2x) \in U$  and  $w = (y, y, 2y) \in W$  for some scalars  $x, y \in \mathbb{F}$ . Their sum is:

$$u + w = (x, -x, 2x) + (y, y, 2y) = (x + y, -x + y, 2x + 2y)$$

Let this resulting vector be  $(a, b, c)$ . We can relate its coordinates to the scalars  $x$  and  $y$ :

$$\begin{aligned} a &= x + y \\ b &= -x + y \\ c &= 2x + 2y = 2(x + y) \end{aligned}$$

Notice that the third coordinate  $c$  is always exactly twice the first coordinate  $a$  (since  $c = 2a$ ).

To ensure that every vector of the form  $(a, b, 2a)$  is indeed in  $U + W$ , we must show that for any given  $a, b \in \mathbb{F}$ , there exist  $x, y \in \mathbb{F}$  that satisfy the system:

$$\begin{aligned} x + y &= a \\ -x + y &= b \end{aligned}$$

So  $y = \frac{a+b}{2}$ ,  $x = \frac{a-b}{2}$ . Because the underlying field  $\mathbb{F}$  is either  $\mathbb{R}$  or  $\mathbb{C}$ , division by 2 is valid, meaning such scalars  $x$  and  $y$  will always exist for any choice of  $a$  and  $b$ .

Therefore, the sum  $U + W$  can be described using symbols as:

$$U + W = \{(a, b, 2a) \in \mathbb{F}^3 : a, b \in \mathbb{F}\}$$

Alternatively, writing the condition as a relationship between coordinates  $(x, y, z)$ :

$$U + W = \{(x, y, z) \in \mathbb{F}^3 : z = 2x\}$$

### Description using no symbols:

The sum is the set of all vectors in the three-dimensional vector space where the third coordinate is exactly twice the first coordinate.  $\square$

**Question 15.** Suppose  $U$  is a subspace of  $V$ . What is  $U + U$ ?

*Proof.* We claim that  $U + U = U$ .

To prove this, we show subset inclusion in both directions.

First, we show that  $U + U \subseteq U$ . Let  $x \in U + U$ . By definition of the sum of subsets,  $x = u_1 + u_2$  for some vectors  $u_1, u_2 \in U$ . Since  $U$  is a subspace, it is closed under vector addition, which implies  $u_1 + u_2 \in U$ . Therefore,  $x \in U$ , establishing that  $U + U \subseteq U$ .

Next, we show that  $U \subseteq U + U$ . Let  $u \in U$ . Since  $U$  is a subspace, it contains the additive identity 0. We can express  $u$  as the sum  $u = u + 0$ . Because  $u \in U$  and  $0 \in U$ , the sum  $u + 0$  is an element of  $U + U$ . Therefore,  $u \in U + U$ , establishing that  $U \subseteq U + U$ .

Since  $U + U \subseteq U$  and  $U \subseteq U + U$ , we conclude that  $U + U = U$ .  $\square$

**Question 16.** Is the operation of addition on the subspaces of  $\mathcal{V}$  commutative? In other words, if  $U$  and  $W$  are subspaces of  $\mathcal{V}$ , is  $U + W = W + U$ ?

*Proof.* Yes, the operation of addition on the subspaces of  $\mathcal{V}$  is commutative.

To prove that  $U + W = W + U$ , we must show that each set is a subset of the other.

First, let  $x \in U + W$ . By the definition of the sum of subspaces,  $x = u + w$  for some vectors  $u \in U$  and  $w \in W$ . Because vector addition in  $V$  is commutative, we have  $u + w = w + u$ . Since  $w \in W$  and  $u \in U$ , the vector  $w + u$  is in  $W + U$ . Therefore,  $x \in W + U$ , establishing that  $U + W \subseteq W + U$ .

Next, let  $y \in W + U$ . By definition,  $y = w + u$  for some vectors  $w \in W$  and  $u \in U$ . Again, by the commutativity of vector addition in  $V$ ,  $w + u = u + w$ . Since  $u \in U$  and  $w \in W$ , the vector  $u + w$  is in  $U + W$ . Therefore,  $y \in U + W$ , establishing that  $W + U \subseteq U + W$ .

Because  $U + W \subseteq W + U$  and  $W + U \subseteq U + W$ , we conclude that  $U + W = W + U$ .  $\square$

**Question 17.** Is the operation of addition on the subspaces of  $\mathcal{V}$  associative? In other words, if  $\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3$  are subspaces of  $\mathcal{V}$ , is

$$(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)?$$

*Proof.* Yes, the operation of addition on the subspaces of  $\mathcal{V}$  is associative.

To prove that  $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$ , we show subset inclusion in both directions.

First, let  $x \in (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$ . By definition,  $x = u + v_3$  for some  $u \in \mathcal{V}_1 + \mathcal{V}_2$  and  $v_3 \in \mathcal{V}_3$ . Since  $u \in \mathcal{V}_1 + \mathcal{V}_2$ , we can write  $u = v_1 + v_2$  for some  $v_1 \in \mathcal{V}_1$  and  $v_2 \in \mathcal{V}_2$ . Substituting for  $u$  yields  $x = (v_1 + v_2) + v_3$ . Because vector addition in  $V$  is associative, we can rewrite this as  $x = v_1 + (v_2 + v_3)$ . Let  $w = v_2 + v_3$ . Since  $v_2 \in \mathcal{V}_2$  and  $v_3 \in \mathcal{V}_3$ , by definition  $w \in \mathcal{V}_2 + \mathcal{V}_3$ . Therefore,  $x = v_1 + w$ , which means  $x \in \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$ . This establishes that  $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 \subseteq \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$ .

Next, let  $y \in \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$ . By definition,  $y = v_1 + w$  for some  $v_1 \in \mathcal{V}_1$  and  $w \in \mathcal{V}_2 + \mathcal{V}_3$ . Since  $w \in \mathcal{V}_2 + \mathcal{V}_3$ , we can write  $w = v_2 + v_3$  for some  $v_2 \in \mathcal{V}_2$  and  $v_3 \in \mathcal{V}_3$ . Substituting for  $w$  yields  $y = v_1 + (v_2 + v_3)$ . By the associativity of vector addition in  $\mathcal{V}$ , we can rewrite this as  $y = (v_1 + v_2) + v_3$ . Let  $u = v_1 + v_2$ . Since  $v_1 \in \mathcal{V}_1$  and  $v_2 \in \mathcal{V}_2$ , by definition  $u \in \mathcal{V}_1 + \mathcal{V}_2$ . Therefore,  $y = u + v_3$ , which means  $y \in (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$ . This establishes that  $\mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3) \subseteq (\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3$ .

Since both sets are subsets of each other,  $(\mathcal{V}_1 + \mathcal{V}_2) + \mathcal{V}_3 = \mathcal{V}_1 + (\mathcal{V}_2 + \mathcal{V}_3)$ .  $\square$

**Question 18.** Does the operation of addition on the subspaces of  $\mathcal{V}$  have an additive identity? Which subspaces have additive inverses?

*Proof.* Yes, the operation of addition on the subspaces of  $\mathcal{V}$  has an additive identity, which is the trivial subspace  $\{0\}$ .

To verify this, let  $U$  be any subspace of  $V$ . By the definition of subspace addition,

$$U + \{0\} = \{u + 0 \mid u \in U\} = \{u \mid u \in U\} = U$$

Therefore,  $\{0\}$  is the additive identity.

Regarding additive inverses, only the trivial subspace  $\{0\}$  has an additive inverse.

To prove this, suppose a subspace  $U$  has an additive inverse  $W$ . By definition, this means  $U + W = \{0\}$ . Because  $W$  is a subspace, it must contain the zero vector  $0$ . Let  $u$  be any vector in  $U$ . The sum  $u + 0$  must be in  $U + W$ . Since  $U + W = \{0\}$ , it follows that  $u + 0 = 0$ , which means  $u = 0$ . Because this must be true for every vector  $u \in U$ , the only element in  $U$  is the zero vector, so  $U = \{0\}$ . Thus, no non-trivial subspace can have an additive inverse.  $\square$

**Question 19.** Prove or give a counterexample: If  $\mathcal{V}_1, \mathcal{V}_2, U$  are subspaces of  $\mathcal{V}$  such that

$$\mathcal{V}_1 + U = \mathcal{V}_2 + U,$$

then  $\mathcal{V}_1 = \mathcal{V}_2$ .

*Proof.* The statement is false. We provide a counterexample.

Let  $V = \mathbb{R}^2$ . Consider the following subspaces:

- $U = \{(x, 0) \mid x \in \mathbb{R}\}$  (the  $x$ -axis)
- $V_1 = \{(0, y) \mid y \in \mathbb{R}\}$  (the  $y$ -axis)
- $V_2 = \{(x, x) \mid x \in \mathbb{R}\}$  (the line  $y = x$ )

First, we evaluate  $\mathcal{V}_1 + U$ . Any vector  $(a, b) \in \mathbb{R}^2$  can be written as  $(0, b) + (a, 0)$ . Since  $(0, b) \in \mathcal{V}_1$  and  $(a, 0) \in U$ , their sum spans the entire space. Thus,  $\mathcal{V}_1 + U = \mathbb{R}^2$ .

Next, we evaluate  $\mathcal{V}_2 + U$ . Any vector  $(a, b) \in \mathbb{R}^2$  can be written as  $(b, b) + (a - b, 0)$ . Since  $(b, b) \in \mathcal{V}_2$  and  $(a - b, 0) \in U$ , this sum also spans the entire space. Thus,  $\mathcal{V}_2 + U = \mathbb{R}^2$ .

We have established that  $\mathcal{V}_1 + U = \mathcal{V}_2 + U = \mathbb{R}^2$ . However,  $\mathcal{V}_1 \neq \mathcal{V}_2$  (for example, the vector  $(0, 1)$  is in  $\mathcal{V}_1$  but not in  $\mathcal{V}_2$ ). Because  $\mathcal{V}_1 + U = \mathcal{V}_2 + U$  does not imply  $\mathcal{V}_1 = \mathcal{V}_2$ , the statement is false.  $\square$

**Question 20.** Suppose

$$U = \{(x, x, y, y) \in \mathbb{F}^4 : x, y \in \mathbb{F}\}.$$

Find a subspace  $W$  of  $\mathbb{F}^4$  such that  $\mathbb{F}^4 = U \oplus W$ .

*Proof.*  $W = \{(0, a, 0, b) \in \mathbb{F}^4 : a, b \in \mathbb{F}\}$ .  $\square$

**Question 21.** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find a subspace  $W$  of  $\mathbb{F}^5$  such that  $\mathbb{F}^5 = U \oplus W$ .

*Proof.*  $W = \{(0, 0, a, b, c) \in \mathbb{F}^5 : a, b, c \in \mathbb{F}\}$ .  $\square$

**Question 22.** Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 : x, y \in \mathbb{F}\}.$$

Find three subspaces  $W_1, W_2, W_3$  of  $\mathbb{F}^5$ , none of which equals  $\{0\}$ , such that  $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$ .

*Proof.*

$$W_1 = \{(0, 0, a, 0, 0) \in \mathbb{F}^5 : a \in \mathbb{F}\}$$

$$W_2 = \{(0, 0, 0, b, 0) \in \mathbb{F}^5 : b \in \mathbb{F}\}$$

$$W_3 = \{(0, 0, 0, 0, c) \in \mathbb{F}^5 : c \in \mathbb{F}\}$$

$\square$

**Question 23.** Prove or give a counterexample: If  $\mathcal{V}_1, \mathcal{V}_2, U$  are subspaces of  $\mathcal{V}$  such that

$$\mathcal{V} = \mathcal{V}_1 \oplus U \quad \text{and} \quad \mathcal{V} = \mathcal{V}_2 \oplus U,$$

then  $\mathcal{V}_1 = \mathcal{V}_2$ .

*Proof.* The statement is false. We provide a counterexample.

- $U = \{(x, 0) \mid x \in \mathbb{R}\}$  (the  $x$ -axis)
- $\mathcal{V}_1 = \{(0, y) \mid y \in \mathbb{R}\}$  (the  $y$ -axis)

- $\mathcal{V}_2 = \{(y, y) \mid y \in \mathbb{R}\}$  (the line  $y = x$ )

□

**Question 24.** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called *even* if

$$f(-x) = f(x)$$

for all  $x \in \mathbb{R}$ . A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is called *odd* if

$$f(-x) = -f(x)$$

for all  $x \in \mathbb{R}$ . Let  $\mathcal{V}_e$  denote the set of real-valued even functions on  $\mathbb{R}$  and let  $\mathcal{V}_o$  denote the set of real-valued odd functions on  $\mathbb{R}$ . Show that  $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$ .

*Proof.* To prove that  $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$ , we must show that  $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$  and  $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$ .

First, we show that  $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$ . Let  $f \in \mathcal{V}_e \cap \mathcal{V}_o$ . Because  $f \in \mathcal{V}_e$ , it must be an even function, so  $f(-x) = f(x)$  for all  $x \in \mathbb{R}$ . Because  $f \in \mathcal{V}_o$ , it must be an odd function, so  $f(-x) = -f(x)$  for all  $x \in \mathbb{R}$ . Equating the two expressions for  $f(-x)$ , we get:

$$f(x) = -f(x)$$

$$2f(x) = 0$$

$$f(x) = 0$$

Since  $f(x) = 0$  for all  $x \in \mathbb{R}$ ,  $f$  is the zero function. Thus,  $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$ .

Next, we show that  $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$ . Let  $f \in \mathbb{R}^{\mathbb{R}}$  be an arbitrary real-valued function. We can rewrite  $f(x)$  identically as:

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

Let  $f_e(x) = \frac{f(x) + f(-x)}{2}$  and  $f_o(x) = \frac{f(x) - f(-x)}{2}$ . We now check if  $f_e \in \mathcal{V}_e$  and  $f_o \in \mathcal{V}_o$ . For  $f_e$ :

$$f_e(-x) = \frac{f(-x) + f(-(-x))}{2} = \frac{f(-x) + f(x)}{2} = f_e(x)$$

Since  $f_e(-x) = f_e(x)$ ,  $f_e$  is an even function, so  $f_e \in \mathcal{V}_e$ . For  $f_o$ :

$$f_o(-x) = \frac{f(-x) - f(-(-x))}{2} = \frac{f(-x) - f(x)}{2} = -\left(\frac{f(x) - f(-x)}{2}\right) = -f_o(x)$$

Since  $f_o(-x) = -f_o(x)$ ,  $f_o$  is an odd function, so  $f_o \in \mathcal{V}_o$ .

Since any arbitrary function  $f \in \mathbb{R}^{\mathbb{R}}$  can be expressed as the sum of an even function  $f_e \in \mathcal{V}_e$  and an odd function  $f_o \in \mathcal{V}_o$ , we have  $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$ .

Since  $\mathcal{V}_e \cap \mathcal{V}_o = \{0\}$  and  $\mathcal{V}_e + \mathcal{V}_o = \mathbb{R}^{\mathbb{R}}$ , we conclude that  $\mathbb{R}^{\mathbb{R}} = \mathcal{V}_e \oplus \mathcal{V}_o$ . □

- **Why is it called Linear Algebra?:**
  - **Linear:** Geometrically, this describes the act of "squishing" or projecting a higher-dimensional space into a lower-dimensional one. Essentially, you can map data from a higher dimensional space to a lower dimensional space, eventually to a linear space. This is the study of how this goal can be achieved.
  - **Algebra:** A vector space over a field can be called an algebra if you define vector multiplication on it.
- **What some other less intuitive fields?:** If you take the integers modulo a prime—like modulo 3, modulo 5, or modulo 7—they form a perfectly valid finite field. If the number you modulo by is not prime (like modulo 4 or 6), it fails to be a field because you get "zero divisors" (e.g.,  $2 \times 3 = 6 \equiv 0 \pmod{6}$ ), meaning you can't divide by 2 or 3).
- **Which came first, linear independence or direct sum?:** Logically it does not matter which comes first because they are the exact same concept applied to two different objects. Linear independence is a property of individual vectors, while a direct sum is a property of entire subspaces.
- **Prove that  $U_1 \cap (U_2 + U_3) = \{0\}$ ,  $U_2 \cap (U_1 + U_3) = \{0\}$ , and  $U_3 \cap (U_1 + U_2) = \{0\}$  is a necessary and sufficient condition for  $U_1 + U_2 + U_3$  to be a direct sum:.**

**Necessity (Forward):** Assume  $U_1 + U_2 + U_3$  is a direct sum. Let  $v \in U_1 \cap (U_2 + U_3)$ . Because  $v \in U_1$ , we can write it as  $v = u_1$  for some  $u_1 \in U_1$ . Because  $v \in U_2 + U_3$ , we can also write it as  $v = u_2 + u_3$  for some  $u_2 \in U_2$  and  $u_3 \in U_3$ . Equating these expressions gives  $u_1 = u_2 + u_3$ , which rearranges to  $u_1 - u_2 - u_3 = 0$ . By the definition of a direct sum, the only way to sum to the zero vector is if each individual component from each subspace is zero. Therefore,  $u_1 = 0$ , meaning  $v = 0$ . This proves  $U_1 \cap (U_2 + U_3) = \{0\}$ . By symmetry, the exact same logic dictates that the other two intersections must also be  $\{0\}$ .

**Sufficiency (Backward):** Assume the three intersection conditions hold, and suppose we have a sum  $u_1 + u_2 + u_3 = 0$ . We can algebraically rearrange this to  $u_1 = -(u_2 + u_3)$ . This shows that  $u_1$  is an element of  $U_1$ , but it is also equal to an element in  $U_2 + U_3$ . Therefore,  $u_1 \in U_1 \cap (U_2 + U_3)$ . By our assumed condition, this intersection contains only  $\{0\}$ , which strictly forces  $u_1 = 0$ . Substituting  $u_1 = 0$  back into our original equation leaves  $u_2 + u_3 = 0$ , or  $u_2 = -u_3$ . This means  $u_2 \in U_2 \cap U_3$ . Since  $U_2 \cap U_3$  is a subset of  $U_2 \cap (U_1 + U_3)$ , which we assumed is  $\{0\}$ ,  $u_2$  must also be 0. Finally, substituting  $u_2 = 0$  forces  $u_3 = 0$ . Because  $u_1 = 0, u_2 = 0$ , and  $u_3 = 0$  is the only solution to  $u_1 + u_2 + u_3 = 0$ , the sum is proven to be direct.

## 2 Finite-Dimensional Vector Spaces

### 2.1 Span and Linear Independence

#### 2.1.1 Linear Combinations and Span

**Definition 2.1** (Linear Combination). A linear combination of a list  $v_1, \dots, v_m$  of vectors in  $V$  is a vector of the form

$$a_1v_1 + \dots + a_mv_m,$$

where  $a_1, \dots, a_m \in \mathbb{F}$ .

**Definition 2.2** (Span). The set of all linear combinations of a list of vectors  $v_1, \dots, v_m$  in  $V$  is called the span of  $v_1, \dots, v_m$ , denoted by  $\text{span}(v_1, \dots, v_m)$ . In other words,

$$\text{span}(v_1, \dots, v_m) = \{a_1v_1 + \dots + a_mv_m : a_1, \dots, a_m \in \mathbb{F}\}.$$

The span of the empty list  $()$  is defined to be  $\{0\}$ .

**Theorem 2.3.** The span of a list of vectors in  $\mathcal{V}$  is the smallest subspace of  $\mathcal{V}$  containing all vectors in the list.

*Proof.* Suppose  $v_1, \dots, v_m$  is a list of vectors in  $\mathcal{V}$ .

First we show that  $\text{span}(v_1, \dots, v_m)$  is a subspace of  $V$ . The additive identity is in  $\text{span}(v_1, \dots, v_m)$  because

$$0 = 0v_1 + \dots + 0v_m.$$

Also,  $\text{span}(v_1, \dots, v_m)$  is closed under addition because

$$(a_1v_1 + \dots + a_mv_m) + (c_1v_1 + \dots + c_mv_m) = (a_1 + c_1)v_1 + \dots + (a_m + c_m)v_m.$$

Furthermore,  $\text{span}(v_1, \dots, v_m)$  is closed under scalar multiplication because

$$\lambda(a_1v_1 + \dots + a_mv_m) = \lambda a_1v_1 + \dots + \lambda a_mv_m.$$

Thus  $\text{span}(v_1, \dots, v_m)$  is a subspace of  $\mathcal{V}$ .

Each  $v_k$  is a linear combination of  $v_1, \dots, v_m$  (to show this, set  $a_k = 1$  and let the other  $a$ 's in equal 0). Thus  $\text{span}(v_1, \dots, v_m)$  contains each  $v_k$ . Conversely, because subspaces are closed under scalar multiplication and addition, every subspace of  $\mathcal{V}$  that contains each  $v_k$  contains  $\text{span}(v_1, \dots, v_m)$ . Thus  $\text{span}(v_1, \dots, v_m)$  is the smallest subspace of  $\mathcal{V}$  containing all the vectors  $v_1, \dots, v_m$ .  $\square$

**Definition 2.4.** A vector space is called finite-dimensional if some list of vectors in it spans the space.

**Definition 2.5** (Polynomial).

- A function  $p : \mathbb{F} \rightarrow \mathbb{F}$  is called a polynomial with coefficients in  $\mathbb{F}$  if there exist  $a_0, \dots, a_m \in \mathbb{F}$  such that

$$p(z) = a_0 + a_1z + a_2z^2 + \dots + a_mz^m$$

for all  $z \in \mathbb{F}$ .

- $\mathcal{P}(\mathbb{F})$  is the set of all polynomials with coefficients in  $\mathbb{F}$ .

**Definition 2.6** (Degree of a Polynomial).

- A polynomial  $p \in \mathcal{P}(\mathbb{F})$  is said to have degree  $m$  if there exist scalars  $a_0, a_1, \dots, a_m \in \mathbb{F}$  with  $a_m \neq 0$  such that for every  $z \in \mathbb{F}$ , we have

$$p(z) = a_0 + a_1z + \dots + a_mz^m.$$

- The polynomial that is identically 0 is said to have degree  $-\infty$ .
- The degree of a polynomial  $p$  is denoted by  $\deg p$ .

### 2.1.2 Linear Independence

**Definition 2.7** (Linearly Independent).

- A list  $v_1, \dots, v_m$  of vectors in  $\mathcal{V}$  is called *linearly independent* if the only choice of  $a_1, \dots, a_m \in \mathbb{F}$  that makes

$$a_1 v_1 + \dots + a_m v_m = 0$$

is  $a_1 = \dots = a_m = 0$ .

- The empty list  $()$  is also declared to be linearly independent.

**Definition 2.8** (Linearly Dependent).

- A list of vectors in  $\mathcal{V}$  is called *linearly dependent* if it is not linearly independent.
- In other words, a list  $v_1, \dots, v_m$  of vectors in  $\mathcal{V}$  is linearly dependent if there exist  $a_1, \dots, a_m \in \mathbb{F}$ , not all 0, such that  $a_1 v_1 + \dots + a_m v_m = 0$ .

**Theorem 2.9** (Linear Dependence Lemma). *Suppose  $v_1, \dots, v_m$  is a linearly dependent list in  $\mathcal{V}$ . Then there exists  $k \in \{1, 2, \dots, m\}$  such that*

$$v_k \in \text{span}(v_1, \dots, v_{k-1}).$$

*Furthermore, if  $k$  satisfies the condition above and the  $k^{\text{th}}$  term is removed from  $v_1, \dots, v_m$ , then the span of the remaining list equals  $\text{span}(v_1, \dots, v_m)$ .*

*Proof.* Because the list  $v_1, \dots, v_m$  is linearly dependent, there exist numbers  $a_1, \dots, a_m \in \mathbb{F}$ , not all 0, such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

Let  $k$  be the largest element of  $\{1, \dots, m\}$  such that  $a_k \neq 0$ . Then

$$v_k = -\frac{a_1}{a_k} v_1 - \dots - \frac{a_{k-1}}{a_k} v_{k-1},$$

which proves that  $v_k \in \text{span}(v_1, \dots, v_{k-1})$ , as desired.

Now suppose  $k$  is any element of  $\{1, \dots, m\}$  such that  $v_k \in \text{span}(v_1, \dots, v_{k-1})$ . Let  $b_1, \dots, b_{k-1} \in \mathbb{F}$  be such that

$$v_k = b_1 v_1 + \dots + b_{k-1} v_{k-1}. \quad (1)$$

Suppose  $u \in \text{span}(v_1, \dots, v_m)$ . Then there exist  $c_1, \dots, c_m \in \mathbb{F}$  such that

$$u = c_1 v_1 + \dots + c_m v_m.$$

In the equation above, we can replace  $v_k$  with the right side of (1), which shows that  $u$  is in the span of the list obtained by removing the  $k^{\text{th}}$  term from  $v_1, \dots, v_m$ . Thus removing the  $k^{\text{th}}$  term of the list  $v_1, \dots, v_m$  does not change the span of the list.  $\square$

**Theorem 2.10.** *In a finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.*

*Proof.* Suppose that  $u_1, \dots, u_m$  is linearly independent in  $\mathcal{V}$ . Suppose also that  $w_1, \dots, w_n$  spans  $\mathcal{V}$ . We need to prove that  $m \leq n$ . We do so through the process described below with  $m$  steps; note that in each step we add one of the  $u$ 's and remove one of the  $w$ 's.

#### Step 1

Let  $B$  be the list  $w_1, \dots, w_n$ , which spans  $\mathcal{V}$ . Adjoining  $u_1$  at the beginning of this list produces

a linearly dependent list (because  $u_1$  can be written as a linear combination of  $w_1, \dots, w_n$ ). In other words, the list

$$u_1, w_1, \dots, w_n$$

is linearly dependent.

Thus by the linear dependence lemma, one of the vectors in the list above is a linear combination of the previous vectors in the list. We know that  $u_1 \neq 0$  because the list  $u_1, \dots, u_m$  is linearly independent. Thus  $u_1$  is not in the span of the previous vectors in the list above (because  $u_1$  is not in  $\{0\}$ , which is the span of the empty list). Hence the linear dependence lemma implies that we can remove one of the  $w$ 's so that the new list  $B$  (of length  $n$ ) consisting of  $u_1$  and the remaining  $w$ 's spans  $\mathcal{V}$ .

**Step  $k$ , for  $k = 2, \dots, m$**

The list  $B$  (of length  $n$ ) from step  $k - 1$  spans  $\mathcal{V}$ . In particular,  $u_k$  is in the span of the list  $B$ . Thus the list of length  $(n + 1)$  obtained by adjoining  $u_k$  to  $B$ , placing it just after  $u_1, \dots, u_{k-1}$ , is linearly dependent. By the linear dependence lemma (2.9), one of the vectors in this list is in the span of the previous ones, and because  $u_1, \dots, u_k$  is linearly independent, this vector cannot be one of the  $u$ 's.

Hence there still must be at least one remaining  $w$  at this step. We can remove from our new list (after adjoining  $u_k$  in the proper place) a  $w$  that is a linear combination of the previous vectors in the list, so that the new list  $B$  (of length  $n$ ) consisting of  $u_1, \dots, u_k$  and the remaining  $w$ 's spans  $\mathcal{V}$ .

After step  $m$ , we have added all the  $u$ 's and the process stops. At each step as we add a  $u$  to  $B$ , the linear dependence lemma implies that there is some  $w$  to remove. Thus there are at least as many  $w$ 's as  $u$ 's.  $\square$

**Theorem 2.11.** *Every subspace of a finite-dimensional vector space is finite-dimensional.*

*Proof.* Suppose  $\mathcal{V}$  is finite-dimensional and  $U$  is a subspace of  $\mathcal{V}$ . We need to prove that  $U$  is finite-dimensional. We do this through the following multistep construction.

**Step 1**

If  $U = \{0\}$ , then  $U$  is finite-dimensional and we are done. If  $U \neq \{0\}$ , then choose a nonzero vector  $u_1 \in U$ .

**Step  $k$**

If  $U = \text{span}(u_1, \dots, u_{k-1})$ , then  $U$  is finite-dimensional and we are done. If  $U \neq \text{span}(u_1, \dots, u_{k-1})$ , then choose a vector  $u_k \in U$  such that

$$u_k \notin \text{span}(u_1, \dots, u_{k-1}).$$

After each step, as long as the process continues, we have constructed a list of vectors such that no vector in this list is in the span of the previous vectors. Thus after each step we have constructed a linearly independent list, by the linear dependence lemma. This linearly independent list cannot be longer than any spanning list of  $\mathcal{V}$ . Thus the process eventually terminates, which means that  $U$  is finite-dimensional.  $\square$

### 2.1.3 Exercise 2A

**Question 1.** Find a list of four distinct vectors in  $\mathbb{F}^3$  whose span equals

$$\{(x, y, z) \in \mathbb{F}^3 : x + y + z = 0\}.$$

*Proof.* Let  $U = \{(x, y, z) \in \mathbb{F}^3 : x + y + z = 0\}$ . Notice that any vector  $(x, y, z)$  in  $U$  must satisfy  $z = -x - y$ . Therefore, we can express any vector in  $U$  as:

$$(x, y, -x - y) = x(1, 0, -1) + y(0, 1, -1)$$

This implies that the list of two vectors  $(1, 0, -1)$  and  $(0, 1, -1)$  spans  $U$ .

To obtain a list of four distinct vectors that span  $U$ , we can simply include these two vectors and add any two other distinct vectors that also lie in  $U$ . We can generate other vectors in  $U$  by choosing arbitrary values for  $x$  and  $y$  and setting  $z = -x - y$ :

- Let  $x = 1, y = 0 \implies v_1 = (1, 0, -1)$
- Let  $x = 0, y = 1 \implies v_2 = (0, 1, -1)$
- Let  $x = 1, y = 1 \implies v_3 = (1, 1, -2)$
- Let  $x = -1, y = 0 \implies v_4 = (-1, 0, 1)$

Since  $v_1$  and  $v_2$  are in our list, the span of  $(v_1, v_2, v_3, v_4)$  contains all linear combinations of  $v_1$  and  $v_2$ , which means it contains all of  $U$ . Furthermore, since  $v_3, v_4 \in U$ , their inclusion does not expand the span beyond  $U$ .

Thus, the span of the list is exactly  $U$ , and a valid list of four distinct vectors is:

$$(1, 0, -1), (0, 1, -1), (1, 1, -2), (-1, 0, 1)$$

□

**Question 2.** Prove or give a counterexample: If  $v_1, v_2, v_3, v_4$  spans  $\mathcal{V}$ , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans  $\mathcal{V}$ .

*Proof.* The statement is true.

**Proof:** Let  $u \in \mathcal{V}$ . Since the list  $v_1, v_2, v_3, v_4$  spans  $\mathcal{V}$ , there exist scalars  $a_1, a_2, a_3, a_4 \in \mathbb{F}$  such that:

$$u = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$$

We must show that  $u$  can be expressed as a linear combination of the vectors in the new list:  $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ .

Notice that we can rewrite each of the original vectors in terms of the new vectors by using telescoping sums:

$$\begin{aligned} v_4 &= v_4 \\ v_3 &= (v_3 - v_4) + v_4 \\ v_2 &= (v_2 - v_3) + (v_3 - v_4) + v_4 \\ v_1 &= (v_1 - v_2) + (v_2 - v_3) + (v_3 - v_4) + v_4 \end{aligned}$$

Substitute these expressions back into the original equation for  $u$ :

$$\begin{aligned} u &= a_1((v_1 - v_2) + (v_2 - v_3) + (v_3 - v_4) + v_4) \\ &\quad + a_2((v_2 - v_3) + (v_3 - v_4) + v_4) \\ &\quad + a_3((v_3 - v_4) + v_4) \\ &\quad + a_4 v_4 \end{aligned}$$

Distributing the scalars and factoring out the new vectors yields:

$$u = a_1(v_1 - v_2) + (a_1 + a_2)(v_2 - v_3) + (a_1 + a_2 + a_3)(v_3 - v_4) + (a_1 + a_2 + a_3 + a_4)v_4$$

Because the field of scalars  $\mathbb{F}$  is closed under addition, the coefficients  $a_1$ ,  $(a_1 + a_2)$ ,  $(a_1 + a_2 + a_3)$ , and  $(a_1 + a_2 + a_3 + a_4)$  are all valid scalars in  $\mathbb{F}$ .

Thus, any arbitrary vector  $u \in \mathcal{V}$  can be written as a linear combination of  $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ , meaning this list also spans  $\mathcal{V}$ . □

**Question 3.** Suppose  $v_1, \dots, v_m$  is a list of vectors in  $\mathcal{V}$ . For  $k \in \{1, \dots, m\}$ , let

$$w_k = v_1 + \dots + v_k.$$

Show that  $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$ .

*Proof.* To show that  $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$ , we must prove that each set is a subset of the other.

**Part 1:**  $\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$

By definition, for each  $k \in \{1, \dots, m\}$ , the vector  $w_k$  is given by:

$$w_k = v_1 + \dots + v_k$$

This shows that every  $w_k$  is a linear combination of the vectors in the list  $v_1, \dots, v_m$ . Since every  $w_k \in \text{span}(v_1, \dots, v_m)$ , and the span is a vector space closed under addition and scalar multiplication, any linear combination of  $w_1, \dots, w_m$  must also belong to  $\text{span}(v_1, \dots, v_m)$ . Therefore,

$$\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m).$$

**Part 2:**  $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$

We can express each vector  $v_k$  as a linear combination of the  $w$  vectors. For  $k = 1$ , we clearly have:

$$v_1 = w_1$$

For any  $k \in \{2, \dots, m\}$ , we can isolate  $v_k$  by subtracting consecutive  $w$  vectors:

$$\begin{aligned} w_k - w_{k-1} &= (v_1 + \dots + v_{k-1} + v_k) - (v_1 + \dots + v_{k-1}) \\ &= v_k \end{aligned}$$

This demonstrates that every  $v_k$  can be written as a linear combination of  $w_1, \dots, w_m$  (specifically,  $w_k - w_{k-1}$ ). Since every  $v_k \in \text{span}(w_1, \dots, w_m)$ , it follows that any linear combination of  $v_1, \dots, v_m$  is also in  $\text{span}(w_1, \dots, w_m)$ . Therefore,

$$\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m).$$

$\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m)$  and  $\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m)$ , the two sets must be equal.  $\square$

**Question 4.**

- (a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.
- (b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

*Proof.*

**Part (a)**

( $\implies$ ) Suppose the list containing only  $v$  is linearly independent. By definition of linear independence, the equation  $av = 0$  implies  $a = 0$  for  $a \in \mathbb{F}$ . If  $v$  were equal to the zero vector ( $v = 0$ ), then any non-zero scalar  $a \in \mathbb{F}$  (such as  $a = 1$ ) would satisfy  $1 \cdot 0 = 0$ . This would mean a non-trivial linear combination equals zero, making the list linearly dependent. This is a contradiction. Therefore,  $v \neq 0$ .

( $\impliedby$ ) Suppose  $v \neq 0$ . Let  $a \in \mathbb{F}$  be a scalar such that  $av = 0$ . Since  $v$  is not the zero vector, it must be that  $a = 0$  (otherwise, we could multiply both sides by the multiplicative

inverse  $1/a$  to get  $v = (1/a)0 = 0$ , contradicting  $v \neq 0$ ). Since  $a = 0$  is the only solution to  $av = 0$ , the list of length one is linearly independent.

**Part (b)**

Let  $v_1, v_2$  be a list of length two in a vector space  $V$ .

( $\implies$ ) Suppose the list  $v_1, v_2$  is linearly independent. We must show that neither vector is a scalar multiple of the other. Assume, for the sake of contradiction, that one is a scalar multiple of the other. If  $v_1 = cv_2$  for some scalar  $c \in \mathbb{F}$ , then we can write:

$$1v_1 - cv_2 = 0$$

Since the coefficient of  $v_1$  is  $1 \neq 0$ , this is a non-trivial linear combination that equals 0, which contradicts the assumption of linear independence. Similarly, if  $v_2 = cv_1$ , then  $-cv_1 + 1v_2 = 0$ , which again contradicts linear independence since the coefficient of  $v_2$  is  $1 \neq 0$ . Thus, neither vector can be a scalar multiple of the other.

( $\impliedby$ ) Suppose neither vector is a scalar multiple of the other. We must show that the list  $v_1, v_2$  is linearly independent. Suppose there exist scalars  $a_1, a_2 \in \mathbb{F}$  such that:

$$a_1v_1 + a_2v_2 = 0$$

If  $a_1 \neq 0$ , we can rearrange the equation and divide by  $a_1$  to isolate  $v_1$ :

$$v_1 = -\frac{a_2}{a_1}v_2$$

This implies that  $v_1$  is a scalar multiple of  $v_2$ , which contradicts our initial assumption. Therefore, we must have  $a_1 = 0$ .

Substituting  $a_1 = 0$  into the original equation gives:

$$a_2v_2 = 0$$

If  $a_2 \neq 0$ , then  $v_2 = 0$ . However, if  $v_2 = 0$ , then  $v_2 = 0v_1$ , which means  $v_2$  is a scalar multiple of  $v_1$ , again contradicting our assumption. Therefore,  $a_2$  must also be 0.

Since the only solution is  $a_1 = 0$  and  $a_2 = 0$ , the list  $v_1, v_2$  is linearly independent.  $\square$

**Question 5.** Find a number  $t$  such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in  $\mathbb{R}^3$ .

*Proof.* We can set up the equation:

$$(5, 9, t) = a(3, 1, 4) + b(2, -3, 5)$$

This yields a system of three linear equations:

$$3a + 2b = 5 \tag{2}$$

$$a - 3b = 9 \tag{3}$$

$$4a + 5b = t \tag{4}$$

Thus, the number is  $t = 2$ .  $\square$

**Question 6.** Show that the list  $(2, 3, 1), (1, -1, 2), (7, 3, c)$  is linearly dependent in  $\mathbb{F}^3$  if and only if  $c = 8$ .

*Proof.* For the list of three vectors to be linearly dependent, the third vector must be a linear combination of the first two.

This means we can set up the equation:

$$a(2, 3, 1) + b(1, -1, 2) = (7, 3, c)$$

This produces a system of three linear equations:

$$2a + b = 7 \tag{5}$$

$$3a - b = 3 \tag{6}$$

$$a + 2b = c \tag{7}$$

We can solve for  $a$  and  $b$  using the first two equations. Adding equation (1) and equation (2) together yields:

$$a = 2b \qquad \qquad \qquad = 3c = 8$$

Because the system is only consistent when  $c = 8$ , the third vector is a linear combination of the first two if and only if  $c = 8$ . Thus, the list is linearly dependent if and only if  $c = 8$ .  $\square$

### Question 7.

- (a) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{R}$ , then the list  $1 + i, 1 - i$  is linearly independent.
- (b) Show that if we think of  $\mathbb{C}$  as a vector space over  $\mathbb{C}$ , then the list  $1 + i, 1 - i$  is linearly dependent.

*Proof. Part (a)* We must show that the only real scalars  $a, b \in \mathbb{R}$  that satisfy the equation

$$a(1 + i) + b(1 - i) = 0$$

are  $a = 0$  and  $b = 0$ .

Expanding the equation, we get:

$$a + ai + b - bi = 0$$

Grouping the real and imaginary parts gives:

$$(a + b) + (a - b)i = 0$$

For a complex number to equal zero, both its real and imaginary parts must be zero. This yields the system of linear equations:

$$a + b = 0$$

$$a - b = 0$$

Which implies  $a = 0$  and  $b = 0$ .

Since the only solution is  $a = 0$  and  $b = 0$ , the list  $1 + i, 1 - i$  is linearly independent over  $\mathbb{R}$ .

### Part (b)

Now suppose  $\mathbb{C}$  is a vector space over  $\mathbb{C}$ .

This allows us to write:

$$1(1 + i) - i(1 - i) = 0$$

Here, our chosen scalars are  $c_1 = 1$  and  $c_2 = -i$ . Because  $c_1 \neq 0$  (and  $c_2 \neq 0$ ), we have a non-trivial linear combination that equals zero.

Therefore, the list  $1 + i, 1 - i$  is linearly dependent over  $\mathbb{C}$ .  $\square$

**Question 8.** Suppose  $v_1, v_2, v_3, v_4$  is linearly independent in  $\mathcal{V}$ . Prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

*Proof.* To prove that the list  $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$  is linearly independent, suppose there exist scalars  $a_1, a_2, a_3, a_4 \in \mathbb{F}$  such that:

$$a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) + a_4v_4 = 0$$

Expanding the terms and regrouping by the vectors  $v_1, v_2, v_3, v_4$  gives:

$$a_1v_1 - a_1v_2 + a_2v_2 - a_2v_3 + a_3v_3 - a_3v_4 + a_4v_4 = 0$$

$$a_1v_1 + (a_2 - a_1)v_2 + (a_3 - a_2)v_3 + (a_4 - a_3)v_4 = 0$$

We are given that the list  $v_1, v_2, v_3, v_4$  is linearly independent. By the definition of linear independence, the only solution to this equation is when all coefficients are equal to zero. This gives us the following system of equations:

$$a_1 = 0$$

$$a_2 - a_1 = 0$$

$$a_3 - a_2 = 0$$

$$a_4 - a_3 = 0$$

Because the only solution to the linear combination is the trivial solution  $a_1 = a_2 = a_3 = a_4 = 0$ , the list  $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$  is also linearly independent.  $\square$

**Question 9.** Prove or give a counterexample: If  $v_1, v_2, \dots, v_m$  is a linearly independent list of vectors in  $\mathcal{V}$ , then

$$5v_1 - 4v_2, v_2, v_3, \dots, v_m$$

is linearly independent.

*Proof.* The statement is true.

Suppose there exist scalars  $a_1, a_2, \dots, a_m \in \mathbb{F}$  such that:

$$a_1(5v_1 - 4v_2) + a_2v_2 + a_3v_3 + \dots + a_mv_m = 0$$

Expanding the terms and regrouping by the vectors  $v_1, v_2, \dots, v_m$  gives:

$$5a_1v_1 - 4a_1v_2 + a_2v_2 + a_3v_3 + \dots + a_mv_m = 0$$

$$5a_1v_1 + (a_2 - 4a_1)v_2 + a_3v_3 + \dots + a_mv_m = 0$$

Because we are given that the list  $v_1, v_2, \dots, v_m$  is linearly independent, the coefficients of each vector in this linear combination must be exactly zero. This produces the following system of equations:

$$5a_1 = 0$$

$$a_2 - 4a_1 = 0$$

$$a_3 = 0$$

$$\vdots$$

$$a_m = 0$$

From the first equation, dividing by 5 yields  $a_1 = 0$ . Substituting  $a_1 = 0$  into the second equation yields  $a_2 - 4(0) = 0$ , which gives  $a_2 = 0$ . The remaining equations directly state that  $a_3 = 0, \dots, a_m = 0$ .

Since the only solution is the trivial solution  $a_1 = a_2 = \dots = a_m = 0$ , the list  $5v_1 - 4v_2, v_2, v_3, \dots, v_m$  is linearly independent.  $\square$

**Question 10.** Prove or give a counterexample: If  $v_1, v_2, \dots, v_m$  is a linearly independent list of vectors in  $\mathcal{V}$  and  $\lambda \in \mathbb{F}$  with  $\lambda \neq 0$ , then  $\lambda v_1, \lambda v_2, \dots, \lambda v_m$  is linearly independent.

*Proof.* The statement is true.

Suppose there exist scalars  $a_1, a_2, \dots, a_m \in \mathbb{F}$  such that:

$$a_1(\lambda v_1) + a_2(\lambda v_2) + \dots + a_m(\lambda v_m) = 0$$

Using the associativity of scalar multiplication in a vector space, we can rewrite this equation as:

$$(\lambda a_1)v_1 + (\lambda a_2)v_2 + \dots + (\lambda a_m)v_m = 0$$

We are given that the list  $v_1, v_2, \dots, v_m$  is linearly independent. By the definition of linear independence, the scalar coefficients of the vectors in this linear combination must all equal zero. This gives:

$$\lambda a_1 = 0, \quad \lambda a_2 = 0, \quad \dots, \quad \lambda a_m = 0$$

We are also given that  $\lambda \neq 0$ . Because  $\lambda$  is a non-zero element of the field  $\mathbb{F}$ , it has a multiplicative inverse,  $\lambda^{-1}$ . Multiplying both sides of each equation by  $\lambda^{-1}$  yields:

$$a_1 = 0, \quad a_2 = 0, \quad \dots, \quad a_m = 0$$

Because the only scalars that satisfy the initial equation are  $a_1 = a_2 = \dots = a_m = 0$ , the list  $\lambda v_1, \lambda v_2, \dots, \lambda v_m$  is linearly independent.  $\square$

**Question 11.** Prove or give a counterexample: If  $v_1, \dots, v_m$  and  $w_1, \dots, w_m$  are linearly independent lists of vectors in  $\mathcal{V}$ , then the list  $v_1 + w_1, \dots, v_m + w_m$  is linearly independent.

*Proof.* The statement is false.

Define  $w_i = -v_i$  for each  $i = 1, \dots, m$ .

First, we verify that  $w_1, \dots, w_m$  is linearly independent. Suppose  $a_1, \dots, a_m \in \mathbb{F}$  are scalars such that:

$$a_1 w_1 + \dots + a_m w_m = 0$$

Substituting  $w_i = -v_i$ , we obtain:

$$-a_1 v_1 - \dots - a_m v_m = 0$$

Since  $v_1, \dots, v_m$  is linearly independent, all coefficients must be zero:  $-a_1 = 0, \dots, -a_m = 0$ , which means  $a_1 = 0, \dots, a_m = 0$ . Thus,  $w_1, \dots, w_m$  is a linearly independent list.

Now, consider the list  $v_1 + w_1, \dots, v_m + w_m$ . For each  $i$ , we have:

$$v_i + w_i = v_i - v_i = 0$$

The new list is exactly  $0, \dots, 0$ . Because this list contains the zero vector, it is linearly dependent. This disproves the statement.  $\square$

**Question 12.** Suppose  $v_1, \dots, v_m$  is linearly independent in  $\mathcal{V}$  and  $w \in \mathcal{V}$ . Prove that if  $v_1 + w, \dots, v_m + w$  is linearly dependent, then  $w \in \text{span}(v_1, \dots, v_m)$ .

*Proof.* Because  $v_1 + w, \dots, v_m + w$  is linearly dependent, there exist scalars  $a_1, \dots, a_m \in \mathbb{F}$ , not all zero, such that

$$a_1(v_1 + w) + \dots + a_m(v_m + w) = 0.$$

Expanding and rearranging the terms, we obtain

$$a_1 v_1 + \dots + a_m v_m + (a_1 + \dots + a_m)w = 0.$$

Let  $c = a_1 + \cdots + a_m$ . If  $c = 0$ , then the equation simplifies to

$$a_1v_1 + \cdots + a_mv_m = 0.$$

Since  $v_1, \dots, v_m$  is linearly independent, this implies that  $a_1 = 0, \dots, a_m = 0$ . However, this contradicts the fact that the scalars  $a_1, \dots, a_m$  are not all zero. Therefore, it must be true that  $c \neq 0$ .

Since  $c \neq 0$ , we can divide by  $c$  and solve for  $w$ :

$$w = -\frac{a_1}{c}v_1 - \cdots - \frac{a_m}{c}v_m.$$

This demonstrates that  $w$  is a linear combination of  $v_1, \dots, v_m$ . Thus,  $w \in \text{span}(v_1, \dots, v_m)$ .  $\square$

**Question 13.** Suppose  $v_1, \dots, v_m$  is linearly independent in  $\mathcal{V}$  and  $w \in \mathcal{V}$ . Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m).$$

*Proof.* First, we prove the forward direction ( $\implies$ ). Suppose  $v_1, \dots, v_m, w$  is linearly independent. Assume, for the sake of contradiction, that  $w \in \text{span}(v_1, \dots, v_m)$ . Then there exist scalars  $a_1, \dots, a_m \in \mathbb{F}$  such that

$$w = a_1v_1 + \cdots + a_mv_m.$$

Rearranging this equation gives

$$a_1v_1 + \cdots + a_mv_m - w = 0.$$

This is a linear combination of  $v_1, \dots, v_m, w$  that equals 0, where the coefficients are not all zero (specifically, the coefficient of  $w$  is  $-1$ ). This contradicts the assumption that  $v_1, \dots, v_m, w$  is linearly independent. Therefore, it must be true that  $w \notin \text{span}(v_1, \dots, v_m)$ .

Next, we prove the backward direction ( $\impliedby$ ). Suppose  $w \notin \text{span}(v_1, \dots, v_m)$ . To show that  $v_1, \dots, v_m, w$  is linearly independent, suppose there exist scalars  $a_1, \dots, a_m, b \in \mathbb{F}$  such that

$$a_1v_1 + \cdots + a_mv_m + bw = 0.$$

If  $b \neq 0$ , we can solve for  $w$ :

$$w = -\frac{a_1}{b}v_1 - \cdots - \frac{a_m}{b}v_m.$$

This would imply  $w \in \text{span}(v_1, \dots, v_m)$ , which contradicts our assumption. Therefore, we must have  $b = 0$ . Substituting  $b = 0$  into our equation yields

$$a_1v_1 + \cdots + a_mv_m = 0.$$

Because  $v_1, \dots, v_m$  is linearly independent, we conclude that  $a_1 = 0, \dots, a_m = 0$ . Since all scalars  $a_1, \dots, a_m, b$  are zero, the list  $v_1, \dots, v_m, w$  is linearly independent.  $\square$

**Question 14.** Suppose  $v_1, \dots, v_m$  is a list of vectors in  $V$ . For  $k \in \{1, \dots, m\}$ , let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list  $v_1, \dots, v_m$  is linearly independent if and only if the list  $w_1, \dots, w_m$  is linearly independent.

*Proof.* First, we prove the forward direction ( $\implies$ ). Suppose  $v_1, \dots, v_m$  is linearly independent. To show that  $w_1, \dots, w_m$  is linearly independent, suppose there exist scalars  $a_1, \dots, a_m \in \mathbb{F}$  such that

$$a_1w_1 + a_2w_2 + \cdots + a_mw_m = 0.$$

Substituting  $w_k = v_1 + \cdots + v_k$  for each  $k$ , we obtain

$$a_1(v_1) + a_2(v_1 + v_2) + \cdots + a_m(v_1 + \cdots + v_m) = 0.$$

Rearranging the terms by factoring out each  $v_k$ , we get

$$(a_1 + a_2 + \cdots + a_m)v_1 + (a_2 + \cdots + a_m)v_2 + \cdots + a_mv_m = 0.$$

Because  $v_1, \dots, v_m$  is linearly independent, all coefficients must be zero:

$$\begin{aligned} a_1 + a_2 + \cdots + a_m &= 0 \\ a_2 + \cdots + a_m &= 0 \\ &\vdots \\ a_m &= 0 \end{aligned}$$

Working from the bottom up,  $a_m = 0$ , which implies  $a_{m-1} = 0$ , and so on, until  $a_1 = 0$ . Since all scalars  $a_1, \dots, a_m$  are zero,  $w_1, \dots, w_m$  is linearly independent.

Next, we prove the backward direction ( $\Leftarrow$ ). Suppose  $w_1, \dots, w_m$  is linearly independent. To show that  $v_1, \dots, v_m$  is linearly independent, suppose there exist scalars  $b_1, \dots, b_m \in \mathbb{F}$  such that

$$b_1v_1 + b_2v_2 + \cdots + b_mv_m = 0.$$

Notice that  $v_1 = w_1$ , and for  $k \geq 2$ ,  $v_k = w_k - w_{k-1}$ . Substituting these expressions into our equation yields

$$b_1(w_1) + b_2(w_2 - w_1) + b_3(w_3 - w_2) + \cdots + b_m(w_m - w_{m-1}) = 0.$$

Rearranging the terms to factor out each  $w_k$ , we obtain

$$(b_1 - b_2)w_1 + (b_2 - b_3)w_2 + \cdots + (b_{m-1} - b_m)w_{m-1} + b_mw_m = 0.$$

Because  $w_1, \dots, w_m$  is linearly independent, all coefficients must be zero:

$$\begin{aligned} b_1 - b_2 &= 0 \\ b_2 - b_3 &= 0 \\ &\vdots \\ b_{m-1} - b_m &= 0 \\ b_m &= 0 \end{aligned}$$

From  $b_m = 0$ , it follows that  $b_{m-1} = 0$ , and continuing this chain gives  $b_1 = b_2 = \cdots = b_m = 0$ . Therefore,  $v_1, \dots, v_m$  is linearly independent.  $\square$

**Question 15.** Explain why there does not exist a list of six polynomials that is linearly independent in  $\mathcal{P}_4(\mathbb{F})$ .

*Proof.* The vector space  $\mathcal{P}_4(\mathbb{F})$  consists of all polynomials with coefficients in  $\mathbb{F}$  of degree at most 4. This space is spanned by the list of polynomials  $1, z, z^2, z^3, z^4$ .

The length of this spanning list is 5. In any finite-dimensional vector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.

Since  $\mathcal{P}_4(\mathbb{F})$  has a spanning list of length 5, any linearly independent list in  $\mathcal{P}_4(\mathbb{F})$  must have a length of at most 5. Therefore, a list of six polynomials in  $\mathcal{P}_4(\mathbb{F})$  cannot be linearly independent.  $\square$

**Question 16.** Explain why no list of four polynomials spans  $\mathcal{P}_4(\mathbb{F})$ .

*Proof.* The list of polynomials  $1, z, z^2, z^3, z^4$  is linearly independent in  $\mathcal{P}_4(\mathbb{F})$ . The length of this linearly independent list is 5.

In any finite-dimensional vector space, the length of every spanning list of vectors is greater than or equal to the length of every linearly independent list of vectors.

Because  $\mathcal{P}_4(\mathbb{F})$  contains a linearly independent list of length 5, any spanning list of  $\mathcal{P}_4(\mathbb{F})$  must have a length of at least 5. Therefore, no list of four polynomials can span the space.  $\square$

**Question 17.** Prove that  $\mathcal{V}$  is infinite-dimensional if and only if there is a sequence  $v_1, v_2, \dots$  of vectors in  $\mathcal{V}$  such that  $v_1, \dots, v_m$  is linearly independent for every positive integer  $m$ .

*Proof.* First, we prove the forward direction ( $\implies$ ). Suppose  $\mathcal{V}$  is infinite-dimensional. We will construct the sequence  $v_1, v_2, \dots$  recursively. Since  $\mathcal{V}$  is infinite-dimensional,  $\mathcal{V} \neq \{0\}$ , so there exists a nonzero vector  $v_1 \in \mathcal{V}$ . The list  $v_1$  is linearly independent. Now, suppose for some positive integer  $m$ , we have chosen  $v_1, \dots, v_m$  such that the list is linearly independent. Because  $\mathcal{V}$  is infinite-dimensional,  $v_1, \dots, v_m$  cannot span  $\mathcal{V}$  (if it did,  $\mathcal{V}$  would be finite-dimensional). Therefore, there exists a vector  $v_{m+1} \in \mathcal{V}$  such that  $v_{m+1} \notin \text{span}(v_1, \dots, v_m)$ . Because  $v_{m+1}$  is not in the span of the linearly independent list  $v_1, \dots, v_m$ , the extended list  $v_1, \dots, v_m, v_{m+1}$  is also linearly independent. By mathematical induction, this process can be continued indefinitely to produce a sequence  $v_1, v_2, \dots$  such that every finite prefix  $v_1, \dots, v_m$  is linearly independent.

Next, we prove the backward direction ( $\impliedby$ ). Suppose there exists a sequence  $v_1, v_2, \dots$  of vectors in  $\mathcal{V}$  such that  $v_1, \dots, v_m$  is linearly independent for every positive integer  $m$ . Assume, for the sake of contradiction, that  $\mathcal{V}$  is finite-dimensional. Then  $\mathcal{V}$  is spanned by some list of finite length, let's call this length  $n$ . The length of any linearly independent list in a finite-dimensional vector space is less than or equal to the length of every spanning list. By our assumption, the list  $v_1, \dots, v_{n+1}$  is linearly independent. The length of this linearly independent list is  $n+1$ , which is strictly greater than the length of the spanning list  $n$ . This is a contradiction. Therefore, our assumption that  $\mathcal{V}$  is finite-dimensional must be false, meaning  $\mathcal{V}$  is infinite-dimensional.  $\square$

**Question 18.** Prove that  $\mathbb{F}^\infty$  is infinite-dimensional.

*Proof.* For each positive integer  $k$ , let  $e_k \in \mathbb{F}^\infty$  be the sequence with 1 in the  $k$ -th position and 0 in all other positions.

Consider an arbitrary positive integer  $m$ . We will show that the list  $e_1, \dots, e_m$  is linearly independent. Suppose there exist scalars  $a_1, \dots, a_m \in \mathbb{F}$  such that

$$a_1 e_1 + \dots + a_m e_m = 0.$$

The left side of this equation evaluates to the sequence  $(a_1, \dots, a_m, 0, 0, \dots)$ , while the right side is the zero sequence  $(0, 0, 0, \dots)$ . Equating the components, we immediately obtain  $a_1 = 0, \dots, a_m = 0$ . Thus, the list  $e_1, \dots, e_m$  is linearly independent.

Because  $\mathbb{F}^\infty$  contains a linearly independent list of length  $m$  for every positive integer  $m$ , it cannot be spanned by any finite list of vectors (the length of a linearly independent list cannot exceed the length of any spanning list). Therefore,  $\mathbb{F}^\infty$  is infinite-dimensional.  $\square$

**Question 19.** Prove that the real vector space of all continuous real-valued functions on the interval  $[0, 1]$  is infinite-dimensional.

*Proof.* Let  $\mathcal{V}$  be the real vector space of all continuous real-valued functions on the interval  $[0, 1]$ . Consider the functions  $f_k \in \mathcal{V}$  defined by  $f_k(x) = x^k$  for  $k = 0, 1, 2, \dots$ .

For any positive integer  $m$ , consider the list of  $m$  functions  $f_0, f_1, \dots, f_{m-1}$ . Suppose there exist real scalars  $a_0, a_1, \dots, a_{m-1}$  such that

$$a_0 f_0 + a_1 f_1 + \dots + a_{m-1} f_{m-1} = 0.$$

This implies that for all  $x \in [0, 1]$ , we have

$$a_0 + a_1x + \cdots + a_{m-1}x^{m-1} = 0.$$

The left side of this equation is a polynomial of degree at most  $m-1$ . A non-zero polynomial of degree at most  $m-1$  can have at most  $m-1$  roots. However, this polynomial equals zero for all infinitely many values of  $x$  in the interval  $[0, 1]$ . The only way this is possible is if the polynomial is the zero polynomial, meaning all its coefficients must be zero:  $a_0 = 0, a_1 = 0, \dots, a_{m-1} = 0$ .

Thus, the list  $f_0, f_1, \dots, f_{m-1}$  is linearly independent. Because  $\mathcal{V}$  contains a linearly independent list of length  $m$  for every positive integer  $m$ ,  $\mathcal{V}$  cannot be spanned by any finite list of vectors. Therefore,  $\mathcal{V}$  is infinite-dimensional.  $\square$

**Question 20.** Suppose  $p_0, p_1, \dots, p_m$  are polynomials in  $\mathcal{P}_m(\mathbb{F})$  such that  $p_k(2) = 0$  for each  $k \in \{0, \dots, m\}$ . Prove that  $p_0, p_1, \dots, p_m$  is not linearly independent in  $\mathcal{P}_m(\mathbb{F})$ .

*Proof.* Because  $p_k(2) = 0$  for each  $k \in \{0, \dots, m\}$ , each polynomial  $p_k$  has a root at 2. Therefore, we can factor out  $(z-2)$  from each polynomial. That is, there exist polynomials  $q_0, q_1, \dots, q_m \in \mathcal{P}_{m-1}(\mathbb{F})$  such that

$$p_k(z) = (z-2)q_k(z)$$

for each  $k \in \{0, \dots, m\}$ .

The vector space  $\mathcal{P}_{m-1}(\mathbb{F})$  is spanned by the list of polynomials  $1, z, \dots, z^{m-1}$ , which has a length of  $m$ . The list  $q_0, q_1, \dots, q_m$  consists of  $m+1$  polynomials in  $\mathcal{P}_{m-1}(\mathbb{F})$ .

A fundamental theorem of linear algebra states that in any finite-dimensional vector space, the length of a linearly independent list cannot exceed the length of any spanning list. Because the list  $q_0, q_1, \dots, q_m$  has length  $m+1$ , which is strictly greater than the length of the spanning list  $m$ , it follows that  $q_0, q_1, \dots, q_m$  must be linearly dependent.

Thus, there exist scalars  $a_0, a_1, \dots, a_m \in \mathbb{F}$ , not all zero, such that

$$a_0q_0 + a_1q_1 + \cdots + a_mq_m = 0.$$

Multiplying both sides of this equation by  $(z-2)$ , we obtain

$$a_0(z-2)q_0 + a_1(z-2)q_1 + \cdots + a_m(z-2)q_m = 0,$$

which simplifies to

$$a_0p_0 + a_1p_1 + \cdots + a_mp_m = 0.$$

Because the scalars  $a_0, a_1, \dots, a_m$  are not all zero, the list  $p_0, p_1, \dots, p_m$  is not linearly independent.  $\square$